

VOLKSWAGEN

AKTIENGESELLSCHAFT



CAD/CAM - data exchange with the Volkswagen Group 2012

Publishing information

Overall responsibility:

Thomas Heilmann

Volkswagen; EKDD/2

Phone: +49 5361 9-81950

E-mail: Thomas.Heilmann@volkswagen.de

Anke Purschke

Volkswagen; EKDD/2

Phone: +49 5361 9-41821

E-mail: Anke.Purschke@volkswagen.de

The English translation is believed to be accurate. In case of discrepancies, the German version is alone authoritative and controlling. Numerical notation according to ISO/IEC Directives, Part 2.

Preface

CAD/CAM data exchange with the Volkswagen Group

These pages provide comprehensive information on the general conditions that apply to partners of the Volkswagen Group in the product emergence process (PEP).

CAD/CAM data exchange is the exchange of product information, with focus on design-model and drawing information, between the CAD/CAM systems of the Volkswagen Group and the CAD/CAM systems of external partners. This information is increasingly supplemented with additional scopes, such as metadata information from administration systems (TDM, EDM, PDM) and product structures. The scope considered here does not include information from the bill of materials-oriented 'technical information' of the Volkswagen Group.

In order to meet the challenges posed by cost efficiency, time-to-market and product quality concerns, the automobile industry employs IS systems and virtual technologies. Considering this initial situation and the increasing links between internationally active automobile manufacturers and globally organized development partners, the descriptive product data is indispensable. The quantity, complexity, and variety of CAD/CAM data exchanged during collaboration between the Volkswagen Group and its partners are increasing continuously. Also on the increase is the number of specifications regulating how this data is handled (methodologies).

Standardization of the technology and processes used is vital if we are to ensure unambiguous exchange channels.

Contents

1. Contract, basic principles, prerequisites.....	7
Development conditions	7
Binding basic standards.....	7
2. CAD/CAM methodology	10
Procedural instructions.....	10
Confidentiality.....	11
3. CATIA V4 standards.....	12
CATIA V4 data checking.....	12
CATIA V4 data modeling in view of CATIA V5.....	13
4. CATIA V5.....	14
5. Creo Elements/Pro (Pro/E).....	17
Creo Elements/Pro tools.....	17
Creo Elements/Pro data quality check	18
6. General specifications for CAD data formats	19
Pilot data exchange	21
Native data	21
Neutral data	21
3-D viewing data.....	22
Plot data	22
PDM sheets	23
Additional information about CAD data.....	23
7. Structure information.....	25
General Information.....	25
Exchanging vehicle data with external partners using DAVOS.....	26
ENGDAT data format	27
8. TDM system in the product emergence process VPM, SAVECAT, XChange	28
VPM.....	28
SAVECAT	29
XChange.....	29
9. Binding nature of CAD model data	31
Significance of CAD model data.....	31
Data quality	31

The English translation is believed to be accurate. In case of discrepancies, the German version is alone authoritative and controlling. Numerical notation according to ISO/IEC Directives, Part 2.

Responsibilities	31
Test procedure	31
10. Data exchange	35
Responsibilities	35
Various options for data exchange	35
Comparison of options	36
Technical pilot data exchange	37
Incoming data check	38
Registration for connection set-up	39
Connection variants and bandwidth	40
Technical prerequisites	40
Data storage devices	41
OFTP	41
Compression/encryption	43
Creation of container files	45
"Pro/Engineer - Wildfire 4.0 HyperKVS containers"	46
"Creo Elements/Pro 5.0 HyperKVS containers"	46
CATIA V5 HyperKVS containers	47
Online use of applications	49
HyperKVS	49
NVS - standard parts accessed by partner companies	50
ONNO online standard texts/other applications in the partner company network	52
11. DMU	54
DMU requirements	54
DMU goals	56
Software tools used	56
12. The Volkswagen Group digital factory	59
HLS XM - the Volkswagen Group's factory planning system	60
Robcad - robot simulation and offline programming	63
Plant Simulation – material flow simulator	65
Process Designer DiKab - process planning in car body manufacture	66
Kate (KVS gate)	68
13. CAD-related applications	70
CAD-related applications - CATIA V4/GII	70
CAD-related applications - CATIA V5/CAA	71

CAD-related applications - Creo Elements/Pro - applications	72
FuBIT	73
CAD-related applications -DoLittle	74
14. Engineering Portal	76
15. Abbreviations	77
16. Address list.....	79

1. Contract, basic principles, prerequisites

Basic contract governing development and design work

Responsibilities and general principles for CAD/CAM data exchange must be defined in the contract governing development and engineering work between the partner and the Volkswagen Group. The contract may be concluded in the form of a so-called framework agreement with individual call-offs or in the form of individual contractual regulations. The department responsible for design engineering is the contact partner for this in all cases.

Development conditions

Group Standard VW 99000 "Overall requirements for service provision within the framework of component development" regulates the relationship between the purchaser and the contractor. The scope of services and responsibilities are determined when the order is placed, on the basis of all prescribed Performance Specifications.

The "Overall requirements for service provision within the framework of component development" include these "General conditions" in their "Applicable documents" section.

Binding basic standards

VW 01014 Engineering drawings; drawing frames and text macros

VW 01050 Technical drawings; scales, lines, hatching, break lines

VW 01052 Engineering drawings; representations
Representation of objects in engineering drawings and other technical documentation.

VW 01054 Engineering drawings; dimensioning
Dimensioning in engineering drawings complies with international standards and has been extended to incorporate special Volkswagen Group requirements.

VW 01055 Reference point system RPS; drawings
Dimensioning/producing/controlling individual parts and assemblies for uniform positioning and ensuring uniform dimensional references.

VW 01057 Statistical tolerance calculation of dimension chains
Applicable to any measurable production characteristic or variables that are interconnected.

VW 01058 Engineering Drawings; lettering
Types of lettering for objects in engineering drawings and other technical documentation.

VW 01059 Requirements for CAD/CAM data

Standard series for creating CAD data (CATIA V4, CATIA V5, Creo Elements/Pro (Pro/E)). On principle, the CAD/CAM data to be delivered must be reproducible and processable in the CAD/CAM systems of the Volkswagen Group process chains.

Observance of the design guideline facilitates reliable CAD/CAM process chains, assists with the digital mock-up, serves to improve internal and external data exchange, forms the basis for CAD/CAM working methods (e.g., drawing preparation, 3-D body design, drawing plants), and automatic further processing of data (plotting, layer transfer, data conversion, data management).

- VW 01059-1 Representation of technical characteristics
- VW 01059-2 Geometric and mathematical data quality
- VW 01059-3 CAD system CATIA V4 for vehicle parts
- VW 01059-3 Supplement 1: Process chain-specific layer assignment
- VW 01059-4 Guidelines for the CAD system CATIA V4 in production
- VW 01059-5 CAD system Creo Elements/Pro
- VW 01059-6 CAD system CATIA V5
- VW 01059-6 Supplement 1: Terms
 - Supplement 2: Electrical cable routing
 - Supplement 3: Process chain adapter, PCA and DMU CADPart
 - Supplement 4: Commercial vehicles

VW 01155 Vehicle supply parts; approval of first supply and changes
This standard specifies the approval of first delivery and changes to vehicle supply parts.

VW 99000 Overall requirements for service provision within the framework of component development.

VW 99000-1 Part 1: Overall requirements for service provision within the framework of component development; planning release.

VW 99000-2 Part 2: Overall requirements for service provision within the framework of component development; procurement release.

VW 99000-3 Part 3: Overall requirements for service provision within the framework of component development; design release (Audi) and engineering release (VW)

VW 99000-4 Part 4: Overall requirements for service provision within the framework of component development; build sample approval (BMG).

VW 91101 Environmental standard for vehicles; vehicle parts, materials, operating fluids; avoidance of hazardous substances

This standard provides information on substances, preparations, and products, the use of which is banned, restricted, or subject to other requirements due to the following regulations:

- laws in countries that are sales markets for vehicles produced by the Volkswagen Group, -
- internal decisions within the Volkswagen Group, irrespective of legal regulations and the location of use.

These and other standards are available from www.vwgroupsupply.com (*Information, tools*) under '[Online Standards](#)' (ONNO).

Note: The user profile must allow access to 'Online Standards.' This is not the default setting. For more detailed information, please consult your in-house administrator.

Further Standards or Performance Specifications referenced in the respective component procurement specifications may also form part of the contract.

Standards	
 Mr. Norbert Wisla	Phone: +49 5361 9-48869 E-mail: norbert.wisla@volkswagen.de

2. CAD/CAM methodology

Methodology guidelines have been defined for the processing of design tasks in the process chains. Part and process chain-specific procedures have thus been generated. The methods are adapted due to the further development and modification of the functionality in the CAD systems employed. This applies particularly to the derivation of drawings and the use of parametric approaches.

Training sessions and workshops based on these application methods are also offered, including the involvement of development partners in some cases.

CAD/CAM methodology	
 Mr. Martin Ahlers	Phone: +49 5361 8-963612 E-mail: martin.ahlers@volkswagen.de
 Mr. Leandro Mouzo (T-Systems)	Phone: +34 93 774-4471 E-mail: extern.leandro.mouzo@seat.es
 Mr. Luboš Patka	Phone: +420 326 81-7316 E-mail: lubos.patka@skoda-auto.cz
 Research & Development Mr. Radek Ježek	Phone: +420 326 81-5328 E-mail: radek.jezek@skoda-auto.cz

Procedural instructions

Specific subject areas or brand- and division-specific conventions are regulated in so-called procedural instructions and/or guidelines in the Volkswagen Group. These documents are available from the appropriate Design Engineering department.

Procedural instructions	
 Mr. Stefan Biernoth	Phone: +49 5361 9-46547 E-mail: stefan.biernoth@volkswagen.de
 Mr. Miguel Montano Ibañez	Phone: +34 93 708-7057 E-mail: miguel.montano@seat.es

Confidentiality

Conclusion of a contract includes the partner's obligation to maintain confidentiality. The development partner must apply for a non-disclosure agreement (relating to CAD/CAM data) prior to commencement of the contract through the Volkswagen contact in the appropriate department.

There are different types of non-disclosure agreement. They are designated as follows:

- General confidentiality with contract conclusion
- Non-disclosure agreement on system use
- Confidentiality for access to Design Engineering premises

On principle, all documentation exchanged with the Volkswagen Group must be treated as confidential. The contractual partner is responsible for observing the non-disclosure regulations. This applies to all employees of the partner company who are involved in the project and employees who have access to information about it - including any commissioned third parties. It must be noted that third parties must be included in the provision of services only after written agreement with Volkswagen AG.

Suitable steps, including organizational measures, e.g., spatial separation of workplaces or supervision of the personnel, must be taken to ensure that only the authorized recipient receives the documents. This rule also applies to transferring documents with the aid of technical communication equipment and systems.

3. CATIA V4 standards

Please note that use of the CATIA V4 CAD system as a design tool was discontinued at Volkswagen as of 2010-01-01.

The CATIA V4 standards for the standardized project file to be used, the PTF states and the general settings naturally apply to a specific release status within the Volkswagen Group and are therefore subject to modifications.

Current information regarding the basic conditions for CATIA V4 applicable in the Volkswagen Group (including download area) can be found at www.vwgroupsupply.com (-Information, -Divisions, -Research & Development, -R&D services) under: 'CATIA':

CATIA V4 standards	
 Mr. Michael Ditschun	Phone: +49 5361 896-3616 E-mail: michael.ditschun@volkswagen.de
 Mr. Leandro Mouzo (T-Systems)	Phone: +34 93 774-4471 E-mail: extern.leandro.mouzo@seat.es
 Mr. Stanislav Sedláček	Phone: +420 326 80-7005 E-mail: stanislav.sedlacek@skoda-auto.cz

CATIA V4 data checking

CATIA V4 design guideline VW 01059-3 "Requirements for CAD/CAM data" must be observed when generating the data. At the same time, data quality must be ensured by using check programs during design engineering. This results in fewer faults and reduces the cost of eliminating faults that do occur. The Volkswagen Group has developed the GII applications ENDCHK for this purpose.

Further information can be found at www.vwgroupsupply.com (-Information, -Tools) under: 'Product data quality'.

- All submitted data must have been OK-ed with ENDCHK.
- Other checking tools are not permitted.

CATIA V4 data modeling in view of CATIA V5

In agreed cases where CATIA V4 modeling is unavoidable, minimum requirements for the use of such data in a CATIA V5 environment must be ensured.

Rules must be strictly adhered to for this purpose. These rules were created by a working group of the German automotive manufacturers together with IBM in the form of an interoperability handbook.

The handbook created by the CEG working group (CATIA user group, with members from the companies Audi, BMW, DaimlerChrysler, Porsche and Volkswagen) is subdivided into two main areas:

- Data integration CATIA V4 -> CATIA V5
- Data integration CATIA V5 -> CATIA V4

On principle, all CATIA V4 models should have the following characteristics for migration to CATIA V5:

- /CLN run without errors
- /CHKTOP should not identify any errors in the topology
- /annd34 for all 2-D models still containing CATIA V3 text and dimensions (DRO elements)
- Set solids to Smart prior to migration
- Solids must undergo a "forced update" prior to migration
- Delete laws
- Resolve links
- Delete unused symbols/details
- Force new Boolean

Conditions for generally applicable standards or recommendations for a CATIA V4 model setup are only possible to a limited extent, since these have different effects in some cases depending on the model type (surface or solid model).

Conditions that are independent of the data type:

- Model dimension must be = 2 000 mm
- Ensure consistency:
 - Positional consistency (see VDA 4955/2, VW 01059-2) is less than 0,02 mm in the data set
 - Tangential consistency (see VDA 4955/2, VW 01059-2) is less than 1° in the data set
 - All segments of 3-D curve traces are positionally consistent in the data set
 - All segments of profiles are tangentially consistent in the data set (if desired)
 - All segments of profiles in the data set are coplanar

No selection of body edges for building SOLID e-primitives. In this case, residual solid fragments remain in CATIA V5, e.g., in thickness building. A contour of an individual primitive that exceeds the creation of a solid boundary is preferred here.

4. CATIA V5

Use of CATIA V5 in new vehicle projects has been adopted in the Bodywork, Electrics/Electronics, and Chassis Engineering Departments. Use of CATIA V5 in vehicle projects implemented with CATIA V4 must be agreed with the appropriate department. COP parts retain their original format and are not migrated.

The following points must be considered when using CATIA V5:

- The software packages approved at Volkswagen containing the current CATIA V5 release, the service pack, and the hot fix level to be used must be obtained through your supporting IT company.
- The Volkswagen settings for CATIA V5, as well as associated documents and data, can be found at www.vwgroupsupply.com (-Information, -Divisions, -Research & Development, -R&D services) under: 'CATIA V5'.
- Interfaces: The interfaces offered in HyperKVS can be found in the HyperKVS documents.
- HyperKVS: It is possible to save CATIA V5 data in HyperKVS. Before saving, the data is placed in quarantine. It is only removed from quarantine once the data has been found to be structurally "OK."
- VPM/DMU: The DMU process is carried out on the basis of CATIA V4 data with VPM. In order to safeguard the process, Standards VW 01059-4 and VW 01059-6 must be observed.
- Applications may be made available to suppliers via <https://servicenet.t-systems.de/>.

Engineering Package (VALIDAT V5 Basic, NTOOL, Netline, OUTGEN*) VTA*) RPS*))

- o VALIDAT V5
- o DoLittle for German-English and English-German drawing texts

Standards: Details on exchanging CATIA V5 data are provided in Standard VW 01059-6, as well as in

Supplement 1 (Glossary)

Supplement 2 (Electrical cable routing)

Supplement 3 (PCA process chain adapter) and

Supplement 4 (Commercial vehicles).

The naming conventions and structures are checked by HyperKVS.

*) as soon as available



CATIA V5 in Design Engineering at Volkswagen

Volkswagen AG Design Engineering works with CATIA V5. However, both CAD systems - CATIA V4 and CATIA V5 - are used in parallel in the development process.

- All new parts of new vehicle projects are designed with CATIA V5 in series-production development.
- The designated structure models must be used (this information is provided via the aforementioned link) and filled with the required information while taking into account Standard VW 01059-6.

Queries regarding content must be clarified with the respective department representatives who placed the order.

CATIA V5 in Toolmaking at Volkswagen

For production, the specifications of Standard VW 01059-6 also apply.

Use of CATIA V4 or CATIA V5 must be agreed with the purchaser.

SKODA



CATIA V5 at Skoda

All new vehicle projects in Body and Chassis Design Engineering and Electrics/Electronics must be developed with CATIA V5.

Currently, Skoda uses CATIA V5 in Production.

SEAT

CATIA V5 at Seat

Currently, Seat uses CATIA V5 in Body and Chassis Engineering.

Electrics/Electronics: All new vehicle projects must be developed with CATIA V5.

Use of CATIA V5 is also possible in the Production division (tool design).

The specifications of VW 01059-6 apply to all CATIA V5 developments, in conjunction with specific departmental methodologies.

The CATIA V5 design guideline must be observed when generating the data. At the same time, data quality must be ensured by using check programs during design engineering. This results in fewer faults and reduces the cost of eliminating faults that do occur. The Volkswagen Group has developed the CAA application VALIDAT for this purpose (see also section "The Volkswagen Group digital factory").

Further information can be found at www.vwgroupsupply.com (-Information, -Tools) under: 'Product data quality'.

- All incoming data must be checked using VALIDAT.
- Other checking tools are not permitted.

CATIA V5	
 Mr. Michael Ditschun	Phone: +49 5361 896-3616 E-mail: michael.ditschun@volkswagen.de
 Mr. Martin Ahlers	Phone: +49 5361 896-3612 E-mail: martin.ahlers@volkswagen.de
 Mr. Leandro Mouzo (T-Systems)	Phone: +34 93 774-4471 E-mail: extern.leandro.mouzo@seat.es
 Mr. Stanislav Sedláček	Phone: +420 326 81-7005 E-mail: stanislav.sedlacek@skoda-auto.cz

5. Creo Elements/Pro (Pro/E)

Creo Elements/Pro standards

The design guideline (Standard VW 01059-5) contains specifications regarding:

- Basic settings (part, assembly, and drawing templates)
- Configuration settings (use of the files config.sup, config.pro, din.dtl)
- Drawing frames to be used (formats)
- Naming conventions
- Model structure (layer, simplified representation)
- Model setup (general principles, parameters used, modeling methods)
- Data quality (formal checks)
- Regulations for designing production equipment

Current information regarding the basic conditions applicable in the Volkswagen Group (including download area) can be found at www.vwgroupsupply.com under: (-Information, -Divisions, -Research and Development, -R&D services, -ProE) 'Creo Elements/Pro':

Creo Elements/Pro standards	
 Mr. Thomas Reineke	Phone: +49 5361 896-3614 E-mail: thomas.reineke@volkswagen.de
 Mr. Francesc Mestres (T-Systems)	Phone: +34 93 774-4470 E-mail: extern.francesc.mestres@seat.es
 T-Systems Czech Republic a.s. Mr. David Solín	Phone: +420 326 71-1410 E-mail: solin@t-systems.cz

Creo Elements/Pro tools

In addition to the Group environment aids, design engineers at both Volkswagen Group and external development partners have so-called toolkit applications at their disposal, which play a major role in simplifying and standardizing the design process. A list of these tools can be found under 'CAD-related applications' (page 70).

Creo Elements/Pro data quality check

It must be ensured that satisfactory data quality is achieved as early as during the design process by observing Standard VW 01059-6. To this end, the ModelCHECK testing tool with the Volkswagen Group configuration must be used during the design process. This results in fewer faults and reduces the cost of eliminating faults that do occur.

A test report must be submitted, in consultation with the data recipient, during data transfer.

ModelCHECK quality criteria:

ModelCHECK checks formal quality criteria

Criteria are derived from process chain requirements

Requirements are documented in Standard VW 01059-5

Requirements are prioritized:



Errors must be corrected without fail



Warnings must be checked and corrected, if necessary

Application help "VW ModelCHECK (for KVS)"

To simplify the use of ModelCHECK, the supplier package includes the "VW ModelCHECK" application aid.

The test report generated by "VW ModelCHECK" corresponds to the log in KVS and can be stored in pdf format, printed, and used as a handover report.

Further information can be found in section 'Binding nature of CAD model data' as well as at www.vwgroupsupply.com under (-Information, -Divisions, -Research and Development) '[Product data quality](#)':

Creo Elements/Pro data quality check	
 Mr. Wolfgang Kunnert	Phone: +49 5361 896-3619 E-mail: wolfgang.kunnert@volkswagen.de
 Mr. Francesc Mestres (T-Systems)	Phone: +34 93 774-4470 E-mail: extern.francesc.mestres@seat.es
 Mr. Jan Dvořák	Phone: +420 326 81-8923 E-mail: jan.dvorak@skoda-auto.cz

6. General specifications for CAD data formats

In principle, there are two possible data formats for CAD/CAM data exchange, depending on the type of cooperation:

- native or
- neutral

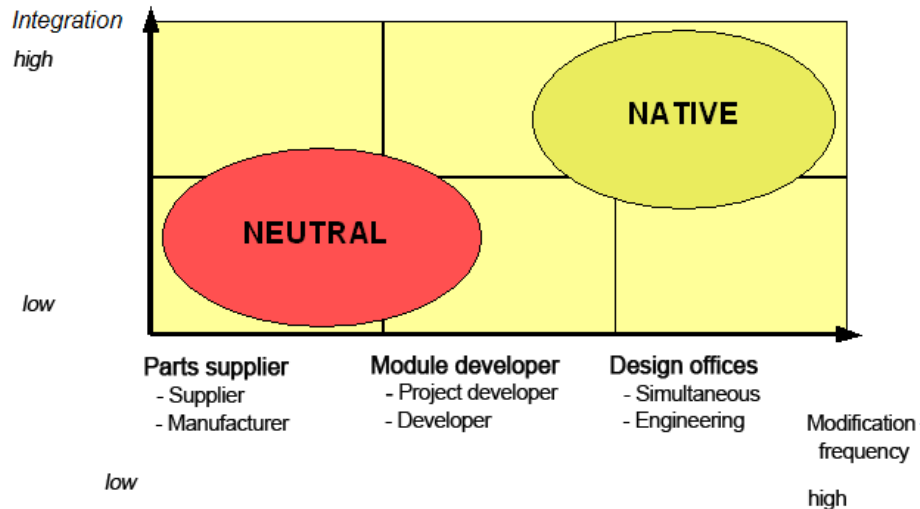


Figure 1: Possible data formats

On principle, the CAD/CAM data to be delivered must be reproducible and processable in the CAD/CAM systems of the Volkswagen Group process chains.

Apart from special-purpose applications, the following CAD/CAM systems are used at the Volkswagen Group as basic systems; for these systems, data can be delivered in the system-specific format:

CATIA V5 Bodywork Engineering

Creo Elements/Pro Powertrain process chain

In specific areas, special systems are used alongside these basic systems. Data exchange in the required format must be agreed upon bilaterally with the respective departments in advance and verified by means of a pilot data exchange.

The following general principles apply in order to prevent supply bottlenecks for Design Engineering, DMU, and other users of CAD data:

No.	Design office/ SE partner/ development supplier	System supplier / parts suppliers
1.	On principle, CAD data must be prepared for the scopes ordered.	
2.	The CAD data includes 3-D surfaces and/or solid geometries (Creo Elements/Pro: only SOLIDS for final model).	
3.	It must be possible to derive the 2-D drawing unambiguously from the surface/solid geometry.	
4.	• CAD data must be prepared in the CAD system of the Volkswagen Group Design Engineering Department that commissioned the work. • CAD data must be delivered in the valid Volkswagen Group application environment.	• CAD data may be prepared in any CAD system. • CAD data must be delivered in the CAD system format(s) of the Volkswagen Group Design Engineering Department that uses the data. • Further processing of the CAD data in the process chain must be verified by means of a pilot data exchange.
5.	If applicable, the additional provision of CATIA V4 file formats must be agreed for data scopes relevant to the DMU. This does not apply to CATIA V5 or Creo Elements/Pro data.	
6.	Starting with a file size of approx. 60 MB (at latest), further processing of the CAD data in the process chain must be verified by means of a pilot data exchange.	
7.	2-D drawings must be delivered in the data format of the 3-D model on which they are based.	
8.	CAD data must comply with the design guidelines and quality requirements of the Volkswagen Group and must be checked accordingly using test programs.	
9.	Details are regulated by this brochure and Standard VW 01059. The currently valid version of this brochure can be found on the B2B supplier platform on the Internet.	
10.	CAD data must be delivered at a time to be defined in the Performance Specification prior to delivery of the physical prototype.	

Pilot data exchange

The pilot data exchange must be carried out prior to the start of the project or before productive data exchange and can take place irrespective of the technical connection setup. It serves the purpose of coordinating the application procedures, system tolerances, and limits. The partner will name a competent contact person.

The results are recorded in development network directives, the most current version of which become an integral part of the contract. Changes in the CAD/CAM systems (versions) or application procedures require a new coordination process. Directives are available in system-specific form and may be obtained from the CAD Interface Laboratory.

Native data

The exchange of native data requires a similarly comprehensive arrangement to that for neutral data formats.

External partners who are integrated into the development process - in terms of data management - in the same way as Volkswagen design engineering departments, must always deliver native data that is created in compliance with the Volkswagen Group design guideline using the CAD system releases and environments currently employed in the Volkswagen Group (i.e., created with CATIA for CATIA and with Creo Elements/Pro for Creo Elements/Pro).

For these heavily integrated partners, creation of data in "foreign" systems for subsequent conversion into the Volkswagen Group system environment is not permitted.

Neutral data

System-neutral CAD/CAM interfaces for the exchange of modifiable geometric information may only be used practically following agreement on the 'common denominator' of system settings and system application methods. The exchange of parametrics or associativities is not possible in this fashion!

For the exchange of neutral, modifiable geometric data, the CAD/CAM contractual partner/supplier must be capable of receiving, processing, and sending at least one of the following neutral formats:

STEP AP214 CC2 (Standard for the Exchange of Product Model Data, ISO 10303-214)

IGES (Initial Graphics Exchange Specification, ANSI standard Y14.26M)

.

Normally, VDAFS (VDA-Flächen-Schnittstelle (VDA surface interface)), version 2.0 without TOP, TMAT, and TLIST, can no longer meet the CAD data exchange requirements between the current versions of the CAD systems. Use of the VDAFS interface may only be agreed upon in special cases where neither STEP nor IGES interfaces are functionally available.

The interfaces are used as follows:

The English translation is believed to be accurate. In case of discrepancies, the German version is alone authoritative and controlling. Numerical notation according to ISO/IEC Directives, Part 2.

STEP AP 214 CC2 for geometric information with topology (wire model, surface model, volume model, and solid) and model structure information

IGES for geometric information (wire model and surface model) and drawing information

Use of suitable interface processors must be agreed upon for generating the neutral data formats.

The Volkswagen Group recommends use of STEP processors. Use of VDAFS is unnecessary, either for quality or for performance reasons.

CAD Interface Laboratory	
 CAD Interface Laboratory Mr. Steffen Hohmann	Phone: +49 5361 9-76589 E-mail: interfacelaboratory@volkswagen.de

3-D viewing data

The additional exchange of simplified data formats is possible for viewing purposes. The Volkswagen Group supports the following data formats in its application environment:

- CreoView
- JT
- VRML (only in exceptional cases).

Plot data

Plot data exchange generally applies to the exchange of drawing information. In most cases, drawings are used for information and archiving purposes. Further processing at the receiving end is frequently not required. Drawing data is generally transmitted in the CAD model or in the form of electronic paper drawings. Vector and raster data provide the opportunity of transferring and reproducing the drawing data via networks without any loss of information. Additionally, raster data forms the basis of many digital drawing archives.

The Volkswagen Group recommends use of raster data as an exchange format for drawings for the purpose of documentation and information.

At the Volkswagen Group, raster drawing data is held in TIFF G4 format. For data exchange, Volkswagen recommends:
 TIFF G4, black/white, 200 dpi, without preview.

In addition, the vector format HPGL/2 is supported.

Plot data

 CAD Interface Laboratory Mr. Steffen Hohmann	Phone: +49 5361 9-76589 E-mail: interfacelaboratory@volkswagen.de
 Mr. Rainer Klinger	Phone: +49 5361 9-76567 E-mail: rainer.klinger@volkswagen.de
 Mr. Xavier Calafí (T-Systems)	Phone: +34 93 773-3297 E-mail: francesc.calafi@seat.es
 Mr. Marek Dvořák	Phone: +420 326 81-8792 E-mail: marek.dvorak@skoda-auto.cz

PDM sheets

A PDM package (product detail mounting instruction) is always created when the mounting/installation of a component or assembly in the vehicle/powertrain assembly line or the pre-assembly of an assembly, e.g., front end module assembly, requires a corresponding representation (see also OA 700).

Either Microsoft Excel or CATIA V5 from Dassault Systems must be used for electronic preparation of the PDM sheets.

The IsoDraw system from PTC (formerly Itedo/version 6.1.04) is still used to work on PDM sheets that were created in ".iso" format.

Creating PDM sheets with IsoDraw is no longer required for future projects. Mandatory requirements, also for the new data storage system in KVS, can be found in the latest "2.1_EKDV_04_PS" process standard on the [Research and Development \(FE\) regulations webpages](#), or may be obtained from the responsible person at Volkswagen.

It is no longer possible to convert PDM sheets that are/were created with IsoDraw into TIFF format.

PDM sheets/(IsoDraw)	
 Ms. Maike Stephan	Phone: +49 5361 9-29306 E-mail: meike.stephan@volkswagen.de
 Mr. Jaume Verdés (T-Systems)	Phone: +34 93 774-4771 E-mail: extern.jaume.verdes@seat.es

Additional information about CAD data

The English translation is believed to be accurate. In case of discrepancies, the German version is alone authoritative and controlling. Numerical notation according to ISO/IEC Directives, Part 2.

Metadata/administrative data

The VDA (German Association of the Automotive Industry) and other standardization committees have defined data exchange protocols to describe the exchange of data with asynchronously connected data-exchange partners. HyperKVS users can use the ENGDAT protocol to send CAD files to partner companies. This data exchange between partner companies and Volkswagen AG takes place within protected networks.

The ENGDAT protocol allows for the combined transfer of several files from a sender to a recipient. Each ENGDAT transfer contains a descriptive abstract file, including information about the sender, recipient, and rudimentary information about the transferred files.

In HyperKVS, each ENGDAT transfer order is automatically accompanied by an additional file in XML format containing further information about the transferred CAD files.

This KVSEXPORT file contains embedded instructions for converting the included meta file into HTML format. Thus, the XML file can be viewed directly with standard HTML browsers, without any further conversion being necessary.

The communication partner therefore has the opportunity to establish a reference between the received CAD files and the HyperKVS metadata. This significantly simplifies communication between partner company employee and Volkswagen design engineer, as the terms used by Volkswagen are known to and can be used by both parties. Use of standard web browser and XML technologies also means that no problems arise in accessing the actual data, which would be presented by other metadata formats.

Besides viewing the metadata, partner companies can of course also evaluate the meta information at the system level with an XML parser and connect the data to their data processing systems.

Comments in the KVSEXPORT files contain further technical information about the web browser compatibility of the embedded XML-HTML conversion and on the use of the XML technology validation.

7. Structure information

CAD/CAM systems in the product emergence process

General Information

Individual component geometries alone are frequently insufficient for the vehicle development process.

Collision-free design of vehicle components also requires their packaging environment. This contains information on the relative positions and relations of components. Such structural information must be made available to the development partner together with the component geometries.

The DAVOS (Datenaustausch von Fahrzeugstrukturen, Data Exchange of Vehicle Structures) software package has been developed for this purpose. DAVOS is an application for providing positioned component scopes; its primary purpose is to supply structural and geometric data in the as-installed position to external development partners with no VPM access.

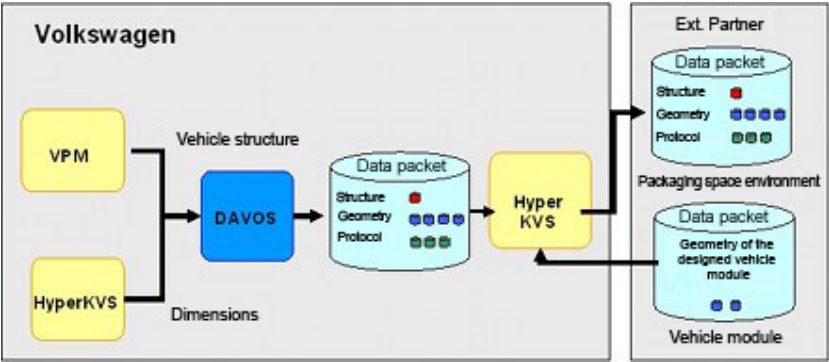
Structural data is provided in an XML file. An HTML file is always supplied for simple visualization. Additionally, CATIA V5, Creo Elements/Pro, JT, or STEP assemblies including the dimensions of all allocated individual parts, may also be supplied. For CATIA V5 assemblies, the component geometry is output in either CATIA V4 or CATIA V5 format; for STEP assemblies, either in CATIA V4 or STEP format.

For audit security reasons, transfer of the result package to external partners may only take place via the persistent KVS system.

Furthermore, DAVOS facilitates updating of already existing results packages. Updating only supplies the differences in the newly loaded/changed geometries plus the complete structure. This keeps the size of the data package to a minimum. A log file is also enclosed, which contains information regarding:

- Deleted components
- Unchanged components
- Newly added components
- Changed components

Exchanging vehicle data with external partners using DAVOS



For the exchange of vehicle packages, the partners involved (sender & receiver) must reach an agreement on the data packages and formats to be exchanged with the CAD and PDM systems involved, including an updating cycle, if required, in a data exchange meeting. The data exchange meeting also decides whether data is exported via the data exchange team at Volkswagen, with the help of a data transfer form, or whether the requesting department compiles the data using DAVOS.

The data exchange meeting is organized by the data exchange team (EKDD/2).

Target group

Potential users are external partners who require positioned vehicle geometries and/or vehicle structures.

DAVOS	
 Data exchange team	Phone: +49 5361 9-78951 E-mail: fe-datenaustausch@volkswagen.de

ENGDAT data format

VDA recommendation 4951 'Remote data transfer of ENGDAT/ENGPART CAD/CAM data' defines the content, structure, and format for

- a machine-readable abstract of CAD/CAM exchange files (engineering data message or ENGDAT message for short) and
- a machine-readable partner profile description (engineering partner data or ENGPART for short)

The ENGDAT message contains necessary information about the respective sender and recipient, format, coding, etc. at the time of transferring the CAD/CAM data and also offers the opportunity of providing the recipient with additional information in text fields.

ENGDAT thus serves as a 'package service,' bundling a number of files with a 'package label' and providing a means for specifying a particular recipient.

As standard, the Volkswagen Group bundles all the aforementioned data formats in one ENGDAT message package.

An additional ENGDAT software component is necessary for ENGDAT data exchange, besides the OFTP software. For more detailed information, please consult your software supplier.

ENGPART import data

An ENGPART file contains information about people and companies, which is relevant to the exchange of personal data between two partners.

This machine-readable ASCII file is exchanged between the partners involved at the beginning of a planned data exchange relationship and also when important changes take place, thereby ensuring a quick automatic update of partner information.

At Volkswagen in Wolfsburg, ENGPART data is imported manually, while at Audi, this takes place semi-automatically.

ENGPART import data

KVS service team

Phone from Germany: 0800 5 877 877

Phone from outside Germany: +49 375 354 28978

E-mail: csn.service@o-s.de

8. TDM system in the product emergence process VPM, SAVECAT, XChange

VPM

General Information

In today's Product Lifecycle Management (PLM) environment, the use of product emergence methods is an essential prerequisite for ensuring competitiveness.

In the CATIA environment, a TDM system in the form of Enovia VPM is available to represent integrated DMU processes. Enovia VPM generates a reusable knowledge base for the entire PLM, thereby ensuring a design without the need for rework.

Fields of application:

- Administration of product structures and relationships between different parts of a product.
- Support for the definition of rule-based configuration and optimization through variant management.
- Forwarding of changes to the relevant project teams and participants responsible for design engineering, development and production.

Use of Enovia VPM in the Volkswagen Group

In the Volkswagen Group, the VPM system is used at all vehicle development locations as well as at several production locations (see list below) for early virtual and geometric verification (DMU). In addition, variant-controlled vehicle structures (such as LHD/RHD, different gearbox/engine combinations, etc.) are generated using HyperKVS documents.

Companies/plants using Enovia VPM:

- Volkswagen AG Passenger Cars (Wolfsburg, Braunschweig, Kassel, Emden, Salzgitter)
- Volkswagen AG Commercial Vehicles (Wolfsburg, Hannover)
- Audi AG
- Volkswagen Sachsen GmbH (Zwickau)
- Skoda Auto a.s.
- Volkswagen de Mexico
- BUGATTI Automobiles S.A.S.
- SEAT S.A.
- Bentley Motors TK
- Volkswagen do Brasil

SAVECAT

General Information

Models from PSN (product structure navigator) or CATIA V4 can be saved on data storage devices as individual models using the SAVECAT 6.0.0 batch program function. Options for the output types are:

- CATIA V4 model
- VOXEL model
- ANSA format
- MEDINA format

The party importing the SAVECAT models does not need a separate import tool (i.e., no SAVECAT either) and the positions from the opened PSN are transferred to the individual models.

Target group

Potential users for writing SAVECAT models:

- All VPM users

Potential users for reading SAVECAT models:

- External partners who do not have VPM
- Teraport users
- Volkswagen departments that need positioned CATIA models for generating non-CATIA formats.

XChange

General Information

The XChange application facilitates data exchange between different Volkswagen VPM entities, and/or the exchange of data between different environments of the same VPM entity.

Both an additive and a synchronized exchange (i.e., taking account of the current state of the target environment) are possible.

In this way, an earlier data transfer may be resumed following any disconnection.

The program allows structures to be exported, including the associated models from VPM via PSN, and then imported at the partner end.

Target group

Potential users for exporting XChange data:

- Designated departments/employees of the Group partners define, export, and transmit XChange data.

Potential users for importing XChange data:

Departments/employees of the Group partners import structures.

VPM	
 Mr. Torsten Pürschel	Phone: +49 5361 896-3839 E-mail: torsten.puerschel@volkswagen.de
 Mr. Guido Fischer	Phone: +49 5361 9-35622 E-mail: guido.fischer@volkswagen.de
 Mr. Raul Gabas Koeppel (T-Systems)	Phone: +34 93 708-7041 E-mail: raul.gabas@seat.es

9. Binding nature of CAD model data

Significance of CAD model data

If 3-D model data is available for a component via CAD, this data always takes precedence over other engineering documents. Drawings always represent only a section of the model data (derived data).

Releases issued by design engineering departments identify the development status of a part. The part is released for a specific application (in the bill of materials). If the term 'release' is used in the context of CAD data, this refers to the assignment of access rights.

On principle, the Volkswagen Group requests provision of CAD data that is as complete as possible. Simple drawing creation is no longer ideal for this process.

Data quality

Volkswagen checks and evaluates the quality of product-defining CAD data with three areas of focus:

- 1) Styling – surface quality and its evaluation by visual inspection
- 2) Organizational (structural) data quality and evaluation
- 3) Geometric mathematical data quality and evaluation

The data quality must meet the standard requirements of the Volkswagen Group. The aim of checking the product data quality is to guarantee usability of the CAD models in all product emergence processes.

Information on this subject can be found at www.vwgroupsupply.com under (-Information, -Divisions, -Research and Development) '[Product data quality](#)'.

Responsibilities

The department responsible for design engineering is also responsible for CAD data.

Data checking must be carried out by the producer of the data in parallel with the design engineering process. The responsible department (data recipient) must at least perform a random check on the supplied data to ensure that the agreed quality criteria have been met.

Test procedure

The data to be checked is the CAD model data generated in the product emergence process through the use of CAD/CAM systems. Apart from the use of internal system tools ('visual inspection'), the software tools ENDCHK and VALIDAT (for CATIA V4), VALIDAT V5 (for CATIA V5), and ModelCHECK (for Creo Elements/Pro) are available for checking purposes.

Agreements regarding criteria and parameters as per Standard VW 01059 and SE checklist (VDA 4961) apply equally to purchaser and contractor.

The creator of the CAD model data uses checking tools during the design process and undertakes to generate and supply CAD model data that

- Complies with design guidelines (ENDCHK, VALIDAT, and ModelCHECK "ok") when delivering CATIA and Creo/Pro data
- Has been checked according to the geometric criteria and parameters of Standard VW 01059 in the case of the delivery of CATIA models In terms of geometry, incoming data is checked in the Volkswagen Group using VALIDAT.

VALIDAT V5 checks both requirements simultaneously.

The department responsible for the design at the data recipient decides whether or not to accept the data. Non-compliance with checking criteria/parameters may result in:

- Acceptance of the data at the responsibility of the data recipient
- Rejection of the data and correction by the creator
- Correction of the data at the expense of the creator (following consultation)

This procedure also applies accordingly to project-specific agreements.

Quarantine

A quarantine concept is applied to specific product data types if models from CATIA V4, CATIA V5 and/or Creo Elements/Pro are stored in KVS. The log contained in the CATIA data checked by the creator is extracted and the test result evaluated. Creo Elements/Pro data is checked with ModelCHECK after it has been stored on the KVS server.

Both CATIA data and Creo Elements/Pro data that do not have the required data quality remain in quarantine.

The following prerequisites must be fulfilled for a CATIA V4 model to be used in the process:

- The product data type selected in the storage dialog corresponds to that used in the ENDCHK run as a directive.
- The ENDCHK status is "OK" (without check result for 'model dimension').
- The ENDCHK version is valid.
- The model dimension 2 000 mm is OK (or an exception to the rule applies.)

The creator of the data bears the cost for any delays in the process brought about by non-compliance with the design guideline.

CATIA V5 data is only released from quarantine once its structural completeness and validity have been determined by structural analysis.

If the following criteria are evident, the data remains in quarantine:

- Files rated as obligatory references are missing from the CATIA V5 container (.tar file).
- CATIA V5 data refers to invalid file formats (e.g., CATIA V4 model, .bmp files).
- The model contains impermissible references (e.g., OLE references [embedded objects such as graphics, Excel spreadsheets, etc.]).
- Components that are not end nodes have been used as structural nodes.
- File names contain impermissible characters.
- CATIA V5 data generated or processed with an ED2 license.
- No root file could be found among the files in the CATIA V5 container.
- CATIA V5 data originates from the PDM system CONNECT.
- File names contains impermissible characters.
- Model generated with incorrect geometric scaling.

The following prerequisites must be fulfilled for CATIA V5 data to be used in the process:

The product data type selected in the storage dialog corresponds to the test profile used for the VALIDAT check.

The VALIDAT status is "OK."

The VALIDAT version is valid.

The creator of the data bears the cost for any delays in the process brought about by non-compliance with the design guideline.

The following preconditions apply when checking Creo Elements/Pro data in HyperKVS and use of the data in the process:

- All imported Creo Elements/Pro data with the product data type TM or TZ are is subjected to checking
- ModelCHECK errors are "not OK" criteria
- All models of a document version must be error-free
- All models are regenerated. The result flows into the total check result (total check result "OK" – "not OK")
- The user receives an e-mail notification if the data is "not OK".
- The check result is documented (check icon, check report, statistics file)
- Data without an "OK" result remains in quarantine unless it is subject to the legacy data regulation or manually removed from quarantine by an authorized person within the context of an emergency release.

CATIA checking tools	
 Mr. Walter Pohlabein	Phone: +49 5361 896-3621 E-mail: walter.pohlabein@volkswagen.de
 Mr. Leandro Mouzo (T-Systems)	Phone: +34 93 774-4471 E-mail: extern.leandro.mouzo@seat.es

ENDCHK, VALIDAT installation support (subject to a fee)	
T-Systems International GmbH	Phone: +49 375 60619-901 E-mail: cax-service@t-systems.com
 T-Systems Czech Republic a.s. Mr. David Solín	Phone: +420 326 71-1410 E-mail: solin@t-systems.cz
 Mr. Leandro Mouzo (T-Systems)	Phone: +34 93 774-4471 E-mail: extern.leandro.mouzo@seat.es

Creo Elements/Pro checking tools (ModelCHECK)	
 Mr. Wolfgang Kunnert	Phone: +49 5361 896-3619 E-mail: wolfgang.kunnert@volkswagen.de
 Mr. Francesc Mestres (T-Systems)	Phone: +34 93 774-4762 E-mail: extern.francesc.mestres@seat.es
 T-Systems Czech Republic a.s. Mr. David Solín	Phone: +420 326 71-1410 E-mail: solin@t-systems.cz
 Mr. Jan Dvořák	Phone: +420 326 81-8923 E-mail: jan.dvorak@skoda-auto.cz

10. Data exchange

Responsibilities

On principle, the responsible Volkswagen Group design engineering department must formulate the request for performing a CAD/CAM data exchange. It must establish agreements concerning the type, content and exchange medium for CAD/CAM data exchange.

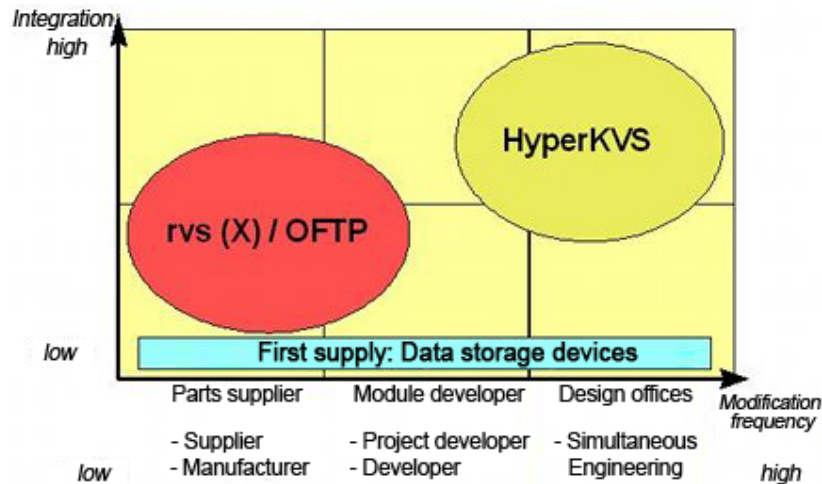


Figure 4: Recommendations for the data exchange

Various options for data exchange

Principle: The exchange of CAD data without accompanying meta information is always informal in nature. Use of the correct versions and statuses must be ensured, particularly when using CAD data for downstream process steps.

The following variants for the exchange of information are supported:

- Data storage devices (asynchronous, offline)
- OFTP (asynchronous, online)
- Online use of applications (synchronous, online)

On principle, the Volkswagen Group recommends use of online solutions for the recurring change service.

Comparison of options

Data storage devices

The technical options must be discussed in advance!

Drawbacks: Time consuming and expensive, since it requires the physical transportation of a medium.

Not auditable. Compatibility and reading problems. Risk of viruses.

Benefits: Possible to supply large volumes of data initially. Data encryption using GnuPG is mandatory.

OFTP/ODETTE with ENGDAT

Drawbacks: Files need to be sent individually if they are not packed using TAR.

Benefits: Fast transfer (depending on the line bandwidth).

Auditable data exchange possible. Data connection is encrypted.

OFTP2/ODETTE with ENGDAT

Drawbacks: Additional expenditure due to key exchange/management, partitioning of the OFTP2 station against internet attacks

Benefits: Rapid transfer of large volumes of data (up to 9 petabytes possible), inexpensive connection, global availability.

Auditable, encrypted data exchange possible as standard without additional components, both for transport and data encryption.

OFTP	
 Mr. Bodo Meier	Phone +49 5361 9-76818 E-mail: bodo.meier@volkswagen.de
 Mr. Daniel Gregor	Phone: +420 326 81-7463 E-mail: daniel.gregor@skoda-auto.cz

Online use of applications

Drawbacks: All functions are carried out in an online session.

Benefits: Synchronous working method provides an optimum working environment with virtually all the options available to design engineers within the Volkswagen Group.

https encryption of the data flow.

Note: Please contact the Operational Services GmbH & Co. KG Service Support Center (SSC) for information about costs arising from acquisition of any hardware/software that may be required for setup or during operation.

A partner must connect to the purchaser's entity. This means: Data intended for Volkswagen must be entered on the Volkswagen entity and data intended for Audi must be entered on the Audi entity.

Technical pilot data exchange

The technical pilot data exchange must be carried out prior to the start of the project or before productive data exchange. Its purpose is to facilitate agreement on the technologies and connections employed.

Technical pilot data exchange	
 CAD Interface Laboratory Mr. Steffen Hohmann	Phone: +49 5361 9-76589 E-mail: interfacelaboratory@volkswagen.de
 Mr. Leandro Mouzo (T-Systems)	Phone: +34 93 774-4471 E-mail: extern.leandro.mouzo@seat.es

Incoming data check

All CAD/CAM data sent to the Volkswagen Group by partner companies undergoes an incoming check. With offline and asynchronous data receipt, the first step is to check for a correct label, which is evaluated if available. Data then undergoes an analysis and preparation process that checks the metadata, file format, and specific aspects of data generated with the aid of computers.

Data acceptance and management

After pilot data exchange has taken place, the relevant design engineering department is responsible for accepting the transmitted data. Successful data exchange is a prerequisite for conclusion of the commissioned work.

No later than conclusion of the design phase, all product-defining data must be stored in the Volkswagen Group's applicable engineering data management system.

'Provision of CAD/CAM data' service of the Volkswagen Group Research & Development Department

External partners that cannot directly access the required CAD/CAM data can obtain it on request via the Volkswagen Research & Development departments; data is then provided by the EKDD department (Design Engineering, Technical Information, Data Exchange, and Cooperation). This data can be provided in native or neutral format. Data conversion to other formats is also possible. The ODETTE File Transfer Protocol (OFTP) is the preferred means of data exchange. In individual cases, data can also be provided on CD or DVD.

You can arrange with the EKDD data exchange team for CAD data to be transferred, provided on data storage devices, converted, and encrypted.

Drawing originals for modification and microfilm enlargements can be ordered from the FE Information Service in EKDD. Procurement of data and drawings from Group subsidiaries may also be arranged for here.

'Provision of CAD/CAM data' service of the Volkswagen Group Research & Development Department

 R&D Data Exchange	Phone: +49 5361 9-78951 E-mail: FE-datenaustausch@volkswagen.de
 R&D Data Provision	Phone: +49 5361 9-29588 E-mail: FE-Informationsservice@volkswagen.de



This regulation does not apply to SEAT. At SEAT, Design Engineering is solely responsible for providing the necessary data, both online via OFTP and offline on tape or CD.



The exchange of CAD data by e-mail is forbidden.

Registration for connection set-up

Operational Services GmbH & Co. KG has been commissioned by Volkswagen and Audi with order processing, operation, and servicing of the data connections.

Details concerning connection to CSN (CAx Supplier Network) for the purpose of data exchange can be found at www.vwgroupsupply.com under (-Information, -Divisions, -Research and Development, -R&D services) 'CAD data exchange':

Both the prerequisites and sequence for connection are described here in detail. Please contact Operational Services GmbH & Co. KG if you need help or have any questions on this subject.



CSN is not used at SEAT or Skoda. The basic technical conditions at SEAT and Skoda are governed by the HyperKVS network access contract.

Registration for connection set-up	
Operational Services GmbH & Co. KG Service Support Center (SSC)	Phone: 0800 5877 877 (Germany) Phone: +49 375 60619-904 (international) Fax: +49 1805 721433-444 E-mail: csn.service@o-s.de
 Mr. Christian Pomar (T-Systems)	Phone: +34 93 773-3151 E-mail: christian.pomar@t-systems.es
 T-Systems Czech Republic, a.s.	Phone: +420 326 71-1444 E-mail: KVS@t-systems.cz
 Mr. David Romero (T-Systems)	Phone: +34 93 773-1712 E-mail: extern.david.romero@seat.es

Connection variants and bandwidth

You need a technical connection in order to exchange data with Volkswagen and Audi. The development partner can choose the type of technical connection from a connection portfolio of various network providers released by Volkswagen and Audi. The following are essential criteria on which the decision for a connection type is based: bandwidth (data transfer rate), network availability (guaranteed network availability, if applicable), access to the data exchange solution from a single workplace (client2site) or a whole network segment of the development partner (site2site).

The Operational Services GmbH & Co. KG Service Support Center offers consulting services to assist in selection of the appropriate variant from the variety of technical connection types.

Technical prerequisites

The technical prerequisites at the partner company vary greatly and are directly dependent on the selected connection variant. Transparent Internet access via any Internet service provider (ISP) is the minimum technical requirement.

Data communication is established on the basis of the TCP/IP protocol. This must be checked and verified by the partner company, especially when OFTP access and the associated OFTP software are used. At present, the following types of access are available to the Volkswagen and Audi partner company network:

- ENX (European Network Exchange), the European automotive industry communication network (e.g., via the provider T-Systems)

The English translation is believed to be accurate. In case of discrepancies, the German version is alone authoritative and controlling. Numerical notation according to ISO/IEC Directives, Part 2.

- VPN (Virtual Private Network), a connection type via the provider Vodafone (only available in Germany)
- VPN Offnet, a connection type via the provider BT (for partner companies outside Germany)
- SFV (standard leased line), e.g., via local providers such as WOBCOM or COM-IN
- Internet-VPN, an inexpensive client2site connection for single workplaces (only for synchronous data exchange via HyperKVS online access; not for OFTP data exchange).

The "CSN initial information" document on the CSN information page of the Operational Services website provides detailed information on the technical connection types:

<http://www.o-s.de/kundenlogin/CSN/>

SKODA



An ENX connection is (currently) not available at Skoda.

Data storage devices

Data may be exchanged by means of CD or DVD in encrypted form (GnuPG). This form of data exchange during the setup period of a data-exchange connection must be agreed with the data recipients in each individual case. Use of offline data storage devices may also be practical, especially for very large data volumes, even after the connection setup, depending on the volume of data to be transferred and the bandwidth available with the line connection.

Connection variants and bandwidth	
Operational Services GmbH & Co. KG Service Support Center (SSC)	Phone: 0800 5877 877 (Germany) Phone: +49 375 60619-904 (international) Fax: +49 1805 721433-444 E-mail: csn.service@o-s.de
 Mr. Christian Pomar (T-Systems)	Phone: +34 93 773-3151 E-mail: christian.pomar@t-systems.es
 Mr. David Romero (T-Systems)	Phone: +34 93 773-1712 E-mail: david.romero@seat.es

OFTP

The English translation is believed to be accurate. In case of discrepancies, the German version is alone authoritative and controlling. Numerical notation according to ISO/IEC Directives, Part 2.

The most common method of data exchange is OFTP with TCP/IP. This is regulated by the Europe-wide ODETTE specifications.

Different applications from various providers are available for this transfer protocol.

OFTP with ENG DAT

ENG DAT (Engineering Data, VDA Recommendation 4951) is a specification for exchanging CAD/CAM file groups, including accompanying information (sender, drawing number, format, coding, etc.). The accompanying information facilitates better monitoring and organization of CAD/CAM data exchange. ENG DAT data exchange consists of transmission of the accompanying message, followed by up to 999 CAD/CAM files. Files are transferred using the OFTP protocol. Nevertheless, the special ENG DAT file naming convention applies.

Parallel to the current OFTP installation, an OFTP2 access is also available now via the internet. The associated data exchange process is currently being set up; if you are interested in this, please contact Mr. Bodo Meier (Volkswagen AG) directly.

An additional ENG DAT software component is necessary for ENG DAT data exchange, besides the OFTP software. For more detailed information, please consult your software supplier.

The Volkswagen Group employs ENG DAT as its standard transfer method, including the proprietary component ComSecure for data encryption.

OFTP	
 Mr. Bodo Meier	Phone +49 5361 9-76818 E-mail: bodo.meier@volkswagen.de
 Mr. Daniel Gregor	Phone: +420 326 81-7463 E-mail: daniel.gregor@skoda-auto.cz

Compression/encryption

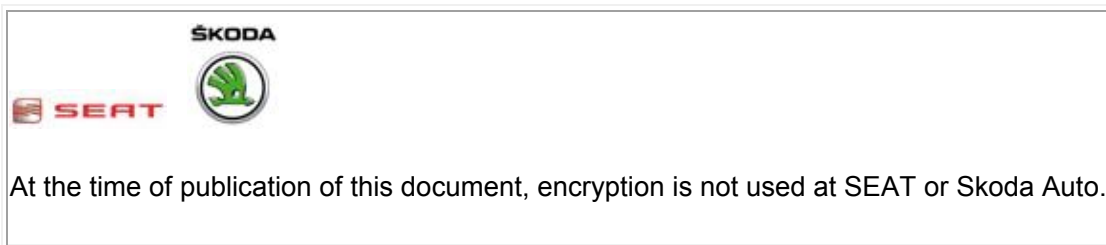
The gzip software package must be used for compression under UNIX. A gzip-version is also available for PCs. The corresponding programs can be downloaded from www.vwgroupsupply.com (-Information, -Divisions, -Research and Development, -R&D services, -CAD data exchange) [Download](#).

If the use of other compression software cannot be avoided, a pilot data exchange is absolutely essential!

Note: The WinZip program can also read and extract .TAR.gz files, if the option 'LF conversion to CR/LF for TAR archives' is deactivated.

Encryption

Only encrypted CAD/CAM data is transferred by means of OFTP or OFTP with ENGDAT. The ComSecure solution is used for encryption, which is available on a wide range of platforms and made available to all partners free of charge. ComSecure is based on established encryption algorithms (RSA and 3DES). Encryption is performed using key pairs certified by the Volkswagen Group. The Operational Services GmbH & Co. KG Service Support Center provides information on key installation and also coordinates key assignment.



E-mail encryption

Volkswagen AG is changing the standard for secure e-mail communication.

Confidential e-mails have previously been sent as encrypted data using S/MIME (end-to-end encryption, content encryption).

In future, e-mails will be transferred via a TLS encrypted connection with a corresponding "company confidential" level of confidentiality.

Volkswagen will offer the following methods as standard technologies for the secure transfer of confidential e-mails, as recommended by the German Association of the Automotive Industry (VDA):

- Transport encryption between e-mail systems (TLS in "mandatory" mode) (preferred)
- E-mail encryption at the encryption gateway by means of PGP or S/MIME

Further information can be found at www.vwgroupsupply.com under (-Information, -Divisions, - Procurement, -IT guidelines)

https://sso.volkswagen.de/portal01/myvw/kbp?path=/content/vwkbc/en/informationen/geschaeftsbereiche/beschaffung/it_richtlinien.html

E-mail encryption	
Operational Services GmbH & Co. KG Automotive Desk Rudolf-Ehrlich-Str. 7 D-08058 Zwickau	Phone: +49 375 60619-903 (Germany) Phone: +49 375 60619-904 (international) Fax: +49 1805 231255 E-mail: servicesupport.center@o-s.de

Creation of container files

Introduction

A container file is a single file consisting of several CAD files bundled together with an additional file containing meta information. This is referred to in the following sections as a container or container file.

The structure and content of the additional file that contains data about other files in the container is defined by specifications of the Volkswagen Group. This file is referred to hereinafter as the meta file.

Rules concerning the structure, creation, and content of container files and the layout of meta files are set out below.

Packaging into containers with GNU-TAR

GNU TAR	
1.	Meta file
2.	CAD file
...	...
n	CAD file

The 'TAR' UNIX tool must be used to bundle CAD files and the meta file. Due to limitations to several TAR implementations, the GNU version of TAR must be used. The first file in the container must be the meta file. During generation of the container with TAR, this is achieved by adding the meta file to the container first.

File name and path conventions for container contents

CAD files must be added to the container without subdirectories. Several Volkswagen standards exist for naming the individual CAD files, which contain applicable specifications for the CAD systems used at Volkswagen AG.

File name and path conventions for the meta file

The meta file must be saved in the container without a path.

The actual file name for the meta file contains several pieces of information intended for further processing, which are separated by the equals sign '='.

The following tables describe the structure of the meta file name. The tables list the information from top to bottom; the actual file name below the table displays the information from left to right.

"Pro/Engineer - Wildfire 4.0 HyperKVS containers"

Value/constant	Description	Example
=	Constant	=
<format designation of root file>	HyperKVS format	WF_ASM TAR
=	Constant	=
<name of application class>		PRO E
=	Constant	=
<name of archived format version>		29
=	Constant	=
<version of application creating archive>		CARL1.0
=	Constant	=
<format version of container format>	Constant	R000
=	Constant	=
<user name>		user
@	Constant	@
<computer name>		abcdef.wob.vw.vwg
=	Constant	=

The following file name for the meta file is derived from the examples in the table (here for a Wildfire 4.0 assembly):

=WF_ASM TAR=PROE=29=CARL1.0=R000=user@abcdef.wob.vw.vwg=

"Creo Elements/Pro 5.0 HyperKVS containers"

Value/constant	Description	Example
=	Constant	=
<format designation of root file>	HyperKVS format	WF_ASM TAR
=	Constant	=
<name of application class>		PRO E
=	Constant	=
<name of archived format version>		31
=	Constant	=

The English translation is believed to be accurate. In case of discrepancies, the German version is alone authoritative and controlling. Numerical notation according to ISO/IEC Directives, Part 2.

The following file name for the meta file is derived from the examples in the table (here for a Wildfire 5.0 assembly):

=WF_ASM TAR=PROE=31=

CATIA V5 HyperKVS containers

Value/constant	Description	Example
=	Constant	=
<format designation of root file>	HyperKVS format	CATPart TAR
=	Constant	=
<name of application class>		CATIA V5
=	Constant	=
<name of archived format version>		R19SP03
=	Constant	=
<time of archive creation>	in YYYYMMDDhhmmss format	20050207145435
=	Constant	=
<version of application creating archive>		KVS-Plugin-402.06-C4[32]
=	Constant	=
<user name>		user
@	Constant	@
<computer name>		abcdef.wob.vw.vwg
=	Constant	=

The following file name for the meta file is derived from the examples in the table:

=CATPart TAR=CATIA V5=R19SP03=20050207145435=KVS-Plugin-402.06-C4[32]=user@abcdef.wob.vw.vwg=

Meta file content and coding

The meta file is an ASCII-coded file (ISO 646) that contains precisely one line of text. This line of text is the file name of the leading model or object: the root of the CAD structure. The line must be concluded with a "line feed" (ASCII code '0x0A').

Due to the coding, umlauts are not permitted in the file name. Furthermore, file names must be selected in accordance with the specifications of Standard VW 01059.

System restrictions for containers

When creating large containers, the possibility of reaching the technical limits of the systems involved must be reckoned with. With the current state of technology, container sizes greater than 2 GByte are problematic. HyperKVS Operations provides information about restrictions to the individual systems.

Compression of containers

Containers may optionally be transferred to HyperKVS in compressed form in order to reduce the amount of storage space, transfer costs, and time involved. The only authorized program for this purpose is the 'gzip' program which originates from the UNIX world.

Note: Users of Microsoft platforms can download the "gzip" program from www.vwgroupsupply.com (-Information, -Divisions, -Research and Development, -R&D services, -CAD data exchange) [Download](#).

Important: The technical limitations of the system regarding the maximum possible size of containers cannot be bypassed through compression, as further processing of containers takes place in uncompressed form!

Handling defective containers

The handling, further processing, and permanent storage of containers that do not meet the formal requirements has not been specified. HyperKVS system behavior based on defective containers which initially might have met with the sending partner company's satisfaction cannot be guaranteed in the longer term.

HyperKVS makes every effort to inform the creator of the data in the case of defective containers.

Explanations

The format designation is the identification generated by HyperKVS format recognition for the known format of a file.

The root file is the file that must be loaded into a CAD system in order to process the complete object.

In HyperKVS systems, data formats are assigned an application class. Such application classes represent the applications with which a particular format can be processed. For example, 'ASM' belongs to the application class Creo Elements/Pro.

The format version of the container format is used for versioning both container and meta file. In a later version, for example, use of subdirectories or conversion of meta files into XML could be introduced. As containers are processed by an automatic program, this is simplified by specifying the version.

Online use of applications

HyperKVS

Most of the features of HyperKVS, which the partner company will be using, can be accessed by means of HTTPS (Hypertext Transfer Protocol Secure). For this purpose, a web browser must be used as the front-end on the client side.

Browser-based access to HyperKVS is via the intranet (a network that cannot be accessed via the internet) or via the partner company network (a network on the intranet that can be accessed by authorized partner companies). It is necessary to encrypt the data flow between client (browser) and server (HyperKVS) to make it secure. Use of appropriate web browsers ensures that the data flow between client and server is always encrypted.

To support the user, a test of whether the software meets these requirements is automatically performed when the HyperKVS homepage is opened. An appropriate message informs the user if the test result is negative.

In order to use HyperKVS, you will need a browser supporting Java script, 128 bit encryption, and HTML version 3 or higher. Furthermore, the browser should support CSS (Cascading Style Sheets). The following browser versions are currently used in the Volkswagen Group:

- Windows operating systems: Internet Explorer version 6.0 SP1 and higher
- UNIX operating systems: IBM AIX, HP-UX, SUN SOLARIS version 1.7 and higher.

HyperKVS: Administrative queries, contract processing, technical connection	
Operational Services GmbH & Co. KG Service Support Center (SSC)	Phone: 0800 5877 877 (Germany) Phone: +49 375 60619-904 (international) Fax: +49 1805 721433-444 E-mail: csn.service@o-s.de
 KVS-SEAT hotline	Phone: +34 93 773-1091 (application) Phone: +34 93 773-3298 (network) E-mail: kvs@seat.es
T-Systems Czech Republic a.s. Hotline	Phone: +420 326 71-1445 Fax: +420 326 71-1420 E-mail: kvs@t-systems.cz

NVS - standard parts accessed by partner companies

Standard parts are essential components of the vehicle. The use of existing standard parts, preferably from the variant reduction for fasteners (VRV) catalog (source: www.vwgrouppsupply.com), saves the work involved in creating a part from scratch. Reducing the number of fastener variants in turn reduces the costs per part and simplifies assembly processes. Standard parts also contribute to a more cost-effective design and a reduction in development times.

With the NVS standard parts management system, the Volkswagen Group provides a convenient information system based on Internet technologies, which is used to obtain information about existing standard parts and installing the standard parts in the current CAD design. This system is also available to the developing partner companies.

The NVS standard parts management system is used to manage standard parts from national, international and Volkswagen Group standards (e.g., screws, bolts, semi-finished products, such as hoses and lines, and consumables, i.e., oils and greases). A distinction is made between geometric, technological, and administrative characteristics. Furthermore, NVS also manages multi-use parts, i.e., parts for which multiple use is highly probable, but whose features deviate in part from standards.

The system supports four research paths:

- Standard number, e.g., DIN EN ISO 14579
- Standard part number, e.g., N 106 411 01
- Classification number, e.g., 61133
- Keyword

If no specific parts are searched for, research via classification number is the more promising method.

Users can download CAD models of the standard parts from HyperKVS to their workstations, in CATIA V4, V5, and Creo Elements/Pro formats. The format to be used is the one used for design and/or assembly. All HyperKVS functions (such as conversion) are available (no conversion of standard parts from CATIA V4 to CATIA V5). All HyperKVS users also have access to NVS.

Introduction to NVS for partner companies at <http://kvspfv1.pfn.vwg:8001/>.

Standard parts assistance (NTA)

Find and use standard parts easily



This tool serves as an aid for CATIA V5 users who want to use standard parts.

Around 1 000 preferred parts have been defined in consultation with the departments within the framework of variant reduction for fasteners (VRV).

These preferred parts are available to all Group employees in the NTA as KVS and CONNECT models for direct installation.

Partner companies can purchase this tool from Cadenas (subject to a fee).

Your contact here is Mr. Blechinger +49 5362 9-48820

Standard parts assistance (NTA) has been developed in cooperation with the Volkswagen AG. DOWNLOAD is possible only at www.vwgroupsupply.com www.vwgroupsupply.com (- Information, -Divisions, -Research and Development, -R&D services, -CAD data exchange).

NVS (standard parts accessible to partner companies)

Operational Services GmbH & Co.KG
Service Support Center (SSC)

Phone 0800 5 877 877 (Germany)
Phone: +49 375 60619-904 (international)
Fax: +49 1805 721433-444
E-mail: csn.service@o-s.de

ONNO online standard texts/other applications in the partner company network

Volkswagen's internal technical rules can be found at www.vwgroupsupply.com under (- Information, -Tools, -Online standard texts) '[Online standard texts](#)' .

Note: The user profile must allow access to 'Standards Online.' This is not the default setting. For more detailed information, please consult your system administrator.

Other applications in the partner company network

Other Volkswagen and Audi applications can be used in particular network segments and by special user groups, depending on access rights and port release. If required, please refer to the contact partner for the applications.

(CAx Supplier Network – CSN) supplied by Volkswagen:

Engineering Portal with single sign-on (SSO) for specified applications	
List of activities	
PEP navigation	ProductEmergenceProcess navigation
RGS online	ReifeGradSpiegel (Maturity Management System) online
WebDMU:	Web-based Digital Mock-Up
HyperKVS	HyperKonstruktionsDatenVerwaltungsSystem (Engineering Data Management System)
ZMB	Zonenmanagement-Berichtssystem (Zone Management Reporting System)
WebTESI	Webbasierte Technische Stücklisten Information (Web-based Technical Bill of Materials Information System)
WebTEIVON	Webbasierte Teile Nummern Verwaltung Online (Web-based Part Number Assignment Online)
PDB	Product DescriptionBook

Engineering data and part information	
ECA	Engineering Center Aggregate (Powertrain Engineering Center)
HyperKVS	Hyper Konstruktionsdaten-VerwaltungsSystem (Engineering Data Management System)
NVS	Normteile-Verwaltungs-System (Standard Parts Management System)
File version management	
CVS	Concurrent Version System

Data management	
CAE Bench	ComputerAidedEngineering data management for numerical simulation

Vehicle information	
CARS benchmark system	Car data Retrieval System
CARS input system	Car data Retrieval System

Production materials	
GAS-TI	Globales Anfrage-System - Technische Information Global tendering system - technical information

Other applications in the partner company network	
 Mr. Heiko Rolof	Phone: +49 5361 9-49737 E-mail: heiko.rolof@volkswagen.de
Operational Services GmbH & Co. KG Service Support Center (SSC)	Phone: 0800 5877 877 (Germany) Phone: +49 375 60619-904 (international) Fax: +49 1805 721433-444 E-mail: csn.service@o-s.de
 Mr. Christian Pomar (T-Systems)	Phone: +34 93 773-3151 Fax: +34 93 773-3252 E-mail: christian.pomar@seat.es
	Phone: +34 93 773-1712

The English translation is believed to be accurate. In case of discrepancies, the German version is alone authoritative and controlling. Numerical notation according to ISO/IEC Directives, Part 2.

Mr. David Romero (T-Systems)	E-mail: david.romero@seat.es
------------------------------	--

The Volkswagen Group regards the provision 3-D data suitable for DMU as an integral part of all development cooperation.

11. DMU

DMU is the abbreviation for Digital Mock-Up.

A mock-up is a scale 1:1 physical model for examining the packaging space of complex assemblies. This procedure is simulated on the computer and hence referred to as a "digital" mock-up.

The Volkswagen Group therefore understands DMU to be a virtual assembly, which is seen as a preliminary stage towards the virtual prototype. It is the complete structural and spatial description of a vehicle in a CA system,

i.e., complete representation of the geometry in 3-D with the associated assembly structures, fasteners, travel spaces, product properties, and administrative information.

This goal is reached using:

- 3-D CAD data (surfaces, volumes, other geometries)
- Kinematic mechanisms
- Static and dynamic envelopes of moving components
- Structural information (bill of materials)
- Administrative attributes (authorizations, status, etc.)

DMU requirements

To ensure consistent use of DMU, the DMU requirements of this guideline must be observed, along with guidelines and work instructions issued by the engineering departments or for each individual project.

Role and objective of DMU in the product emergence process

At each milestone, the DMU zone manager reports the current product status in the DMU to the project management.

Releases may only be granted when compatibility with the current environment has been verified in the DMU.

Requirements for 3-D CAD data for DMU verification

In order to keep the development DMU up-to-date, working versions of 3-D CAD data are required as follows:

- P = package data (packaging space dummies)
- K = concept data

The English translation is believed to be accurate. In case of discrepancies, the German version is alone authoritative and controlling. Numerical notation according to ISO/IEC Directives, Part 2.

- D = production data

This is to ensure that every component can be checked in the DMU with respect to its current environment before the agreed release date, so that the actual release can be assured.

Package data:

Simple volume models and placeholders that include the required packaging space

Concept data:

Volume or surface models that include the technical function and connection (fastener)

Production data:

Complete volume or surface models that describe production parts (as per VW 01059)

The contractor, in its role as developer or development supplier, must provide the required 3-D CAD data to the purchaser's design engineer (or the work statuses for examinations at short notice in the current environment) and report the degree of maturity of the CAD data as well as the application (e.g., DMU data freeze, concept data) by the deadlines agreed with the DMU zone manager/module coordinator (= data freeze for DMU zone evaluation in the context of active DMU zone management) or on call if applicable.

Envelopes

For moving and movable components, the components' travel ranges must be provided in the form of dynamic geometric envelopes in 3-D CAD in consultation with the purchaser's design engineer. It must be ensured that the agreed envelopes correspond to the configured DMU vehicle structure. So that envelopes do not have to be changed completely in the event of changes, they must be subdivided into reasonable sub-assemblies.

Product data type THD or THS must be used for CAD data administration.

Current DMU environment at the developer or development supplier

In order to observe the current test environment at the purchaser from the development DMU, the contractor may request the required data from the purchaser.

Assembly and disassembly simulations using DMU

Production equipment, including fasteners, must be considered during DMU examinations.

Production equipment is provided in the form of 3-D data in the HyperKVS system by the purchaser's Manufacturing Planning department.

Component positioning

Procedural instruction 5.FS-2-001 "CAD Data compliance" applies to Audi; the following regulation applies to all other Group companies:

For components that are used at several locations in the vehicle and as such have to be positioned several times (e.g., radar), data exchange between contractor and purchaser is performed using a transformation matrix (data for component positions).

This also applies to COPs (e.g., switches) that have to be transferred from the as-designed position into the vehicle position (for DMU examination).

The contractor provides the complete information concerning the data position. Accordingly, when the contractor transfers data to the purchaser, the positioning information must be

stored separately from the geometry. Product data type TMP ("Teilmodell Positionsinformation = part model positioning information) is used for storage in HyperKVS. The TMP data set must contain the coordinate axes of the installation positions for each vehicle.

The TMP files are stored under the part number in HyperKVS (in the design status link, the TMP is linked with the associated part model (TM) to form the design status).

Variants are distinguished by means of vehicle alternatives.

The TMP is used to position the components in the Virtual Product Model (VPM) (use MOVE to move the model in CATIA). The positioned model is stored in the PSN tree of the VPM in the correct positional arrangement.

DMU goals

The goal of using DMU is to safeguard the development process by means of virtual installation and simulation of components and assemblies before they are actually manufactured.

Efforts are being made to reduce development time and costs, while at the same time raising product quality through:

- Optimizing cooperation between departments and development partners in DMU meetings
- Early representation and simulation of the finished product before its actual production
- Early fault detection and avoidance
- Early influencing of the product by Manufacturing and After-Sales Service
- Quick and competent evaluation of changes

Software tools used

Visualization

Volkswagen expects the delivery of CATIA V5 data with structures resulting from correctly applied methods. The data must be archived in HyperKVS. The whole vehicle DMU is currently still realized based on CATIA V4 data sets (VPM). Volkswagen in-house processes are used for conversion to CATIA V4. Some departments work on the basis of Creo Elements/Pro.

DMU Navigator and CreoView are the visualization and analysis software packages used in the Group for the DMU process.

CONNECT based on TEAMCENTER is currently being introduced in the Group as the new PDM system. Accompanying this, the visualization tool VISMOCKUP will be introduced as the successor to DMU-Navigator.

Structural information

HyperKVS is the administrative basis for all CAD data in the Volkswagen Group. It is the central data hub of the Group and serves both the internal and worldwide exchange of data.

For the DMU process in the Group, the CAD data from HyperKVS is available in structured assemblies and positionally correct in the Team Data Managers.

The CATIA area uses the VPM (Virtual Product Modeller) and CONNECT as its structure management system, as well as Windchill for the Creo Elements/Pro environment.

Web-based applications are currently being developed in order to optimize data exchange between different hardware platforms and simplify access to data and data structures for external partners. To this end, it will be possible for partner companies to access this data and these structures directly via secure channels.

DMU structure information and DMU data can be accessed within the CSN (at Audi AG via the Engineering Portal).

The current DMU process

The general rules for data exchange with the Volkswagen Group, as specified in the other chapters of this document, apply.

The data must be delivered as 3-D CAD models in HyperKVS, with the appropriate product data type, depending on their use (PDAs: EN, TM, TMZ, TMP, TMA, TMD, TMS, TMU, TZ, THD, THS, ECP, EKR, ELV) together with:

- Part number and part designation
- Design release number and HyperKVS version date
- Name and telephone number of the design engineer
- Organizational unit, organization name, and engineering group

Only the substructure 'Alternative' may be used! No second substructure or one by any other name is allowed. Product data types that are not governed by the design status, such as EN, must be stored in HyperKVS with status "I" in an auditable manner.

On principle, 3-D data must be delivered with positioning information with regard to the vehicle position of the respective vehicle project. Symmetrically opposite parts must be represented with their own data record and part number.

For moving components, the static 3-D description must be supplemented by any finished kinematic mechanisms and the travel ranges of the components in the form of dynamic geometric envelopes.

Departments that use the Creo Elements/Pro CAD system require the data in Creo Elements/Pro format. The Creo Elements/Pro data is reprocessed for the vehicle DMU in a Volkswagen Group-internal process.

It is beneficial for cooperation if the data is provided in the form of intermediate versions, so that the usability of the developed components and assemblies in the vehicle position may be continuously checked within the Volkswagen Group.

Development partners are furthermore requested to consult the design guidelines for further details prior to data exchange and to clarify these with their design engineering contact.

DMU	
	Phone: +49 5361 9-38933

The English translation is believed to be accurate. In case of discrepancies, the German version is alone authoritative and controlling. Numerical notation according to ISO/IEC Directives, Part 2.

Mr. Christian Kutscha	E-mail: christian.kutscha@volkswagen.de
 Mr. Blas Asensio Torres	Phone: +34 93 708-7983 E-mail: blas.asensio@seat.es
 Mr. Martin Kmínek	Phone: +420 326 81-5646 E-mail: martin.kminek@skoda-auto.cz

12. The Volkswagen Group digital factory

Digital factory is used in the automotive industry as a generic term for an extensive network of digital models and methods, including process planning and 3-D visualization. Its purpose is the integrated planning, realization, control, and continuous improvement of all essential factory processes in connection with the product.

A vision has been developed with the following core statements for the 'digital factory' project currently being undertaken in the Volkswagen Group:

- The digital factory portrays all logistics and manufacturing processes that characterize the actual factory.
- The actual factory is only built once the number of parts and quality have been verified in the digital factory.
- The digital factory is a model of the actual factory at all times in the product life span.

The digital factory is essentially based on two pillars:

- Early influencing of the product from a manufacturing perspective
- Safeguarding planning and optimizing the manufacturing and logistics process

The methods of the digital factory are already being used more and more in the planning and design engineering departments in all phases of planning.

Partner companies and suppliers, as well as planning companies, design offices, and system suppliers, are now integrated in the product emergence process (PEP) in all phases of planning. This is conditional upon partner companies using the software tools and application methods employed in the Volkswagen Group.

The following sections describe a selection of different solutions used in the digital factory, which have defined procedures in place and are in productive use in vehicle projects in which partner companies are integrated.

Digital factory	
 Mr. Frank Jelic	Phone: +49 5361 9-37862 E-mail: frank.jelic@volkswagen.de
 Mr. Uwe Koch	Phone: +49 5361 9-40644 E-mail: uwe.koch2@volkswagen.de
 Mr. Jürgen Kunhenn	Phone: +49 5361 9-39577 E-mail: juergen.kunhenn@volkswagen.de
 Mr. Jordi Zapatero Baro	Phone: +34 93 708-7176 E-mail: jordi.zapatero@seat.es
 Mr. Leoš Cervený	Phone: +420 326 81-9439 E-mail: leos.cerveny@skoda-auto.cz

HLS XM - the Volkswagen Group's factory planning system

HLS is a client/server solution for company-wide management of documents, facilities, and inventories in factory planning. Its modular concept and use of the latest integration technologies mean that it can adapt perfectly to the business processes of the company. The multitude of migration and integration solutions mean that HLS can be used in all phases of factory planning. Web access is also available for information purposes.

The main focuses of the HLS application are:

- Hall layout planning
- Building planning
- Office planning
- Architectural engineering (TGA)
- Materials handling systems
- Paint shop planning
- Facilities management

Description of the HLS system landscape

For nearly all brands and locations of the Volkswagen Group, the HLS standard package consists of the MicroStation CAD system, the ProjectWise project management software, and applications based on MicroStation. The database system employed for HLS is ORACLE.

The following software versions are used in HLS:

The English translation is believed to be accurate. In case of discrepancies, the German version is alone authoritative and controlling. Numerical notation according to ISO/IEC Directives, Part 2.

-
- ProjektWise Vers. 08.11.09.122
 - MicroStation Vers. 08.11.07.446
 - Venturis Vers. 10.2
 - Architecture Vers. 08.11.07.87
 - Bentley Facilities Planner Vers. 08.11.09.65

CAD data formats:

The MicroStation DGN format is the only graphical data format used in HLS.

Information for partner companies

All available layout and factory model documents are managed and maintained in the HLS factory planning system. The data required for exporting these documents must be extracted only from the data stock of HLS in MicroStation native format or from software applications based on MicroStation.

The project team members from Volkswagen, Audi, and other brands and locations are responsible for providing external planning partners/suppliers with all required data in this manner.

Data checking:

Currently, CAD files received from the contractor are subjected to a visual inspection for errors by the responsible planner when they are imported into the HLS management system. Implementation of automatic checking is planned for 2013. This includes

- the DGN data format as well as
- compliance with the specified HLS drawing guidelines.

If the check is negative, the data cannot be integrated into HLS.

Applicable documents, guidelines, and standards:

- Product Development Specification for data exchange with the Volkswagen Group's HLS factory planning system
- The document can be found at www.vwgroupsupply.com under (-Information, -Divisions, -Research and Development, -R&D services, -CAD data exchange, -Relevant documents, -CAD/CAM data exchange, -The Volkswagen Group digital factory) [here](#).

Electronic data exchange

In 2009, the ASIC (Application System Inter Connection) data exchange system was introduced for use in HLS.

ASIC is a communication system for the asynchronous transfer of large files. It is a client/server environment comprising a single ASIC server with which any number of clients can communicate. Communication is always initiated by the client, which must have unrestricted internet access. HTTPS is used as the transfer protocol, thereby providing an encrypted connection. ASIC's special feature is the upload/download manager. Even if data packets are lost during transfer or the connection breaks down completely, the process can restart at the same position as soon as the connection between ASIC client and server is restored. The data transfer volume is thus kept as low as possible. Checksums are also used to check whether the file to be transferred has completely reached the target address.

The necessary information is provided below:

At www.vwgroupsupply.com under *(-Information, -Divisions, -Research and Development, -R&D services, -CAD data exchange, -Relevant documents, -CAD/CAM data exchange, -The Volkswagen Group digital factory)*.

- Guideline
- Installation and configuration instructions
- ASIC front end user manual
- as well as the installation package

HLS contacts

The purchaser's organizational unit is responsible for the project workflow. The member of the organizational unit responsible for the HLS project is the contact person for the contracted planning partners and suppliers in all matters relating to CAD order processing.

Contacts:

HLS	
 User support HLS hotline	Phone: +49 5361 9-46192
 Design Engineering Mr. Horst Röhl	Phone: +49 5361 896-3930 E-mail: horst.roehl@volkswagen.de
 Volkswagen support Mr. Rainer Koslik	Phone: +49 5361 9-20977 E-mail: rainer.koslik@volkswagen.de
 ASIC support Dr. Ommo Ommen	Phone: +49 5361 9-25082 E-mail: extern.ommo.ommen@volkswagen.de
 Skoda support Mr. Leoš Červený	Phone: +420 326 81-9439 E-mail: leos.cerveny@skoda-auto.cz
 SEAT support Mr. Jordi Zapatero Baro	Phone: +34 93 708-7176 E-mail: jordi.zapatero@seat.es

Robcad - robot simulation and offline programming

Robcad is a 3-D system for planning, simulation, optimization, and offline programming of automated and manual manufacturing equipment. It is an important element in the digital factory.

Robcad (process simulation) is a licensed software package from Siemens/PLM.

The most important modules employed in the Volkswagen Group are:

- eM-Spot for welding applications
- eM-Paint for painting applications
- eM-Human for ergonomic studies
- eM-OLP for creating robot programs, depending on the robot make, and
- Downloading into the robot control unit

The English translation is believed to be accurate. In case of discrepancies, the German version is alone authoritative and controlling. Numerical notation according to ISO/IEC Directives, Part 2.

Range of services

Robcad is used for the following tasks:

- Realistic modeling of manufacturing cells
- Investigating the accessibility of operating points
- Investigating collision avoidance
- Adherence to cycle times, ergonomics, surface
- Creating motion sequences and paths
- Offline creation of robot programs
- Workplace design/ergonomics

Volkswagen Group applications:

The Volkswagen Group has its own custom software that models Volkswagen Group-specific robot control units in Robcad for offline programming with Robcad-OLP. This is the VKRC2/4 controller software for Kuka robots, VRS1 for Volkswagen Group standard robots, and VRJ3 for Fanuc robots.

Information for partner companies

More detailed information about the versions currently employed in the Volkswagen Group and Volkswagen Group standards, such as instructions for partner companies, can be found at <http://www.vwgroupsupply.com> under: *(-Information, -Divisions, -Research and Development, -R&D services, -CAD data exchange, -Robcad, -Download)*. 'eM Workplace'.

Volkswagen Group-specific robot controller software and robot libraries required for Volkswagen Group orders will be provided by the Volkswagen Group.

Supported platforms

- Windows 7
- Windows XP

Robcad robot simulation and offline programming	
 Mr. Manfred Labersweiler	Phone +49 5361 9-39481 E-mail: manfred.labersweiler@volkswagen.de
 Mr. Jordi Zapatero Baro	Phone: +34 93 708-7176 E-mail: jordi.zapatero@seat.es
 Mr. Leoš Červený	Phone: +420 326 81-9439 E-mail: leos.cerveny@skoda-auto.cz

Plant Simulation – material flow simulator

Plant Simulation (formerly eM-Plant) from Siemens/PLM is a standard software package for object-oriented, graphic, and integrated modeling, simulation, animation, and optimization of various types of processes (e.g., manufacturing, logistics, and business processes). Plant Simulation is the standard tool for preparing and performing process simulations in the Volkswagen Group.

Range of services

Plant Simulation is mainly used in the Volkswagen Group for modeling materials flows during the planning phase and subsequent optimization of the plant at the operator's site. This allows basic materials handling conditions to be modeled, buffer sizes to be optimized, and the logistics and manufacturing concept to be improved in the early planning phases. It also allows statements to be made about the future availability of the plant, its throughput, and restrictions to be expected with regard to variants.

In order to standardize model creation with the Plant Simulation standard product, the VDA Automotive module set is used in the Volkswagen Group. This offers the following advantages:

- Uniform requirements of all OEMs with respect to suppliers/service providers
- Improved legibility of models
- Reusability of models
- Integration of various sub-models created by suppliers or internal simulation experts to form an overall model
- Simplified maintenance of the simulation models.

An availability database (VDB) developed by the Volkswagen Group is used exclusively in the body planning area for parameterization of the models. This database permits the consistent and general definition of basic conditions for model availability.

Use of FlowEval is recommended for pre-parameterization of buffers and buffer zones for simulation with Plant Simulation. FlowEval is an analytic tool for calculating buffer sizes in convergent production structures.

Information for partner companies

The instructions for body-in-white simulation, the "Performance Specification for Materials Flow Simulation," the current version of the "EDP Performance Specification", and the VDA "Instructions for process simulation in the automotive and automotive supplier industries" apply. These documents must be obtained from the appropriate department before the start of the project.

The versions released for use must be agreed with the appropriate department before the start of the project. The VDA Automotive module set must be used as standard. Use of a partner company's own modules is not permitted. If this is unavoidable, the sources of these modules must be submitted to the appropriate departments.

Applicable documents, guidelines, and standards:

- "Performance Specification for Materials Flow Simulation"
- 'EDP Performance Specification' "Instructions for process simulation in the automotive and automotive supplier industries"

The current version is provided by the purchaser or may be requested from the contact persons specified below.

The documents, guidelines and standards are provided in vwgroupsupply, under (-Information, -Tools)

Plant Simulation	
 Mr. Matthias Schwarz	Phone: +49 5361 9-72520 E-mail: matthias.schwarz@volkswagen.de
 Mr. Carsten Pöge	Phone: +49 5361 9-39327 E-mail: carsten.poege@volkswagen.de
 Mr. Leoš Červený	Phone: +420 326 81-9439 E-mail: leos.cerveny@skoda-auto.cz
 Mr. Carlos Banus Giro	Phone: +34 93 708-5058 E-mail: carlos.banus@seat.es

Process Designer DiKab - process planning in car body manufacture

Process Designer DiKab is a digital solution for planning manufacturing processes in car body manufacture.

The English translation is believed to be accurate. In case of discrepancies, the German version is alone authoritative and controlling. Numerical notation according to ISO/IEC Directives, Part 2.

It is primarily used to model manufacturing processes in car body manufacture planning, to design production systems in a 3-D environment, and as a process planning tool for planning, standardization, verification, and evaluation of manufacturing and logistics processes in the product emergence process for car bodies by means of system and robot simulation.

Process Designer has been developed from eM-Planner; it has an improved and enhanced graphic interface and functionality, for example for graphical layout planning. Originally developed as a module of eM-Planner, Process Designer is a stand-alone product and therefore marketed by UGS/Tecnomatix as eM-Designer as of version 7.5.

Goals of process planning

The objective is to establish a central information hub for all technical data in car body manufacture planning. This hub is specifically designed for working with suppliers and partner companies. The main advantage lies in simplified communication and transparency of collaboration between the Volkswagen Group and its suppliers.

Synergies are generated here, above all through simplified reuse or modification of existing planning versions.

Additional applications developed by the Volkswagen Group, such as the automatic generation of manufacturing details (V-plans) and foundation plans on the basis of detailed plant layouts, increase the efficiency of planning new production lines and collaboration with our Group partners.

Use of libraries containing standard production equipment:

Plant planning resources encompass a comprehensive production equipment library of over 1 000 typical Volkswagen and Audi components.

Use and further development of these standard libraries will significantly reduce planning times and expenditures.

The libraries are made available free of charge to all suppliers for use in eM-Planner/Process Designer. Each piece of production equipment is present in the Process Designer/Robcad and HLS (MicroStation) CAD format for quick integration into the planning process.

Users in the Volkswagen Group:

Users in the Volkswagen Group are Car Body Manufacture Planning, Logistics, and the Fixture and Toolmaking Department of the Volkswagen, Volkswagen do Brazil, Audi, Skoda, and SEAT brands. The Performance Specifications for current projects stipulate use of the system by systems suppliers involved in these projects.

If you have any questions concerning eM-Planner and its use in car body manufacture planning, please contact:

Process Designer	
 Mr. Uwe Zaretske	Phone: +49 5361 9-196584 E-mail: uwe.zaretske@volkswagen.de
 Mr. Gunner Jensen	Phone: +49 5361 9 40768 E-mail: Gunner.Jensen@volkswagen.de
 Mr. Leoš Červený	Phone: +420 326 81-9439 E-mail: leos.cerveny@skoda-auto.cz
 Mr. Jordi Zapatero Baro	Phone: +34 93 708-7176 E-mail: jordi.zapaterp@seat.es

Kate (KVS gate)

In HyperKVS, product data are not organized according to the bill of materials but individually by part number.

Without KATE, part numbers in a bill of materials must be retrieved individually by downloading and converted, if necessary.

KATE automates the downloading of all information stored in HyperKVS, which belongs to an existing or personally created bill of materials.

Thanks to various automatic procedures, relevant document versions and formats can be selected and retrieved from HyperKVS. The system also offers a facility for automatic downloading of the latest version of the respective document. Once data retrieval is complete, KATE creates a file which can be imported into downstream tools, such as the Process Designer process planning tool from Siemens/PLM.

Functional scope

- Automated, bill-of-materials-related data retrieval from HyperKVS
- Generation of an eM-Planner bill of materials with associated 3-D information
- Updating of Process Designer product data from HyperKVS
- Definition of individual bills of materials (based on Excel) as input files

Information for partner companies

Currently, Kate is mainly used in the area of manufacturing planning and in combination with the eM-Planner. Kate can be obtained from the department at the Volkswagen Group placing the order.

System requirements

Operating system:

Windows XP, 7

Miscellaneous:

HyperKVS access

Kate	
 Mr. Heimo Lütge	Phone: +49 5361 611-39622 E-mail: Heimo.Luetge@volkswagen.de
 Mr. Jordi Zapatero Baro	Phone: +34 93 708-7176 E-mail: jordi.zapatero@seat.es
 Mr. Leos Červený	Phone: +420 326 81-9439 E-mail: leos.cerveny@skoda-auto.cz

13. CAD-related applications

CAD-related applications -CATIA V4/GII

Please note that use of the CATIA V4 CAD system was discontinued at Volkswagen as of 2010-01-01.

Information on the following CATIA extensions is available via the website

<https://servicenet.t-systems.de/product-design>

of our sales partner T-Systems Enterprise Services GmbH:

- GII basic package ENDCHK, NETLINE, RAMBES, IUA macro collection for CATIA V4
- **VALIDAT** Design data testing program as per VDA 4955
(at www.vwgroupsupply.com under -Information, -Tools, -ProductDataQuality, VALIDAT)
- **DoLittle** is an application integrated into the respective CAD environment that facilitates online creation of bilingual drawings by the EKDD/5 translation team. For further information, see below or www.vwgroupsupply.com under (-Information, -Divisions, -Research and Development, -R&D services, -CAD data exchange, -Relevant documents, -CAD/CAM data exchange, -DoLittle) [here](#).
- **GII weld spots** (SWPII, SWPIII), GII application for the CATIA V4 CAD system for defining fasteners in body design engineering.



The TEACAT function is used at SEAT for text output in CATIA for drawing frames. TEACAT belongs to the GII basic package sold by T-Systems ITC Iberia S.A.U.

The DoLittle function is not used at SEAT.

CATIA V4/GII



T-Systems International GmbH

Phone: +49 375 60619-901

E-mail: cax-service@t-systems.com



Mr. Miguel Delgado (T-Systems)

Phone: +34 93 773-3250

E-mail: extern.miguel.delgado2@seat.es

CAD-related applications - CATIA V5/CAA

The following extensions are available via the website <https://servicenet.t-systems.de> of our sales partner T-Systems Enterprise Services GmbH:

- **Engineering Package** contains all the basic tools required to save CATIA V5 data in HyperKVS and to check for compliance with VW 01059-6: VALIDAT Basic V5, NTOOL, and the HyperKVS plug-in. Furthermore, it is planned to add other applications to the Engineering Package, as soon as they are available, including:
 - VALIDAT V5 Basic – data quality validation as per VW 01059-6
 - NTOOL – profile-controlled naming as per VW 01059-6
 - KVS-Plugin – uploading of CATIA V5 data into HyperKVS
 - NETLINE – generation of 2-D vehicle grids
 - VTA – joining technology application
 - RPS – reference point system
 - LTA – hole technology application
 - OUTGEN – application for deriving process chain adapters
 - **VALIDAT V5** Professional test scopes extending beyond VW 01059-6 and healing support
 - **VW-METHO** method planning applications
 - **DoLittle** is an application integrated into the respective CAD environment that facilitates online creation of bilingual drawings by the EKDD/5 translation team. See below for further information.
 - **VEM** fasteners methodology consisting of a VB macro (Volkswagen only)
 - **BoM2KSL** generation of production equipment BOMs directly from CAD documents in CATIA V5

More detailed information can be found via the aforementioned link.



The DoLittle function is not used at SEAT.

CATIA V5/CAA	
 T-Systems International GmbH	Phone: +49 180 572 143 3510 E-mail: cax-service@t-systems.com
 Mr. Miguel Delgado (T-Systems)	Phone: +34 93 773-3250 E-mail: extern.miguel.delgado2@seat.es

CAVA CATIA Automotive Vehicle Architecture

CAVA is to check the homologation of vehicles for conformity with international standards. It contains the following modules:

- o OVA (Overall Vehicle Architecture)
- o Vision
- o Manikin
- o Safety
- o Toolbox

The programs GKV, LKV, PUPPE, GURT and INSI known from CATIA V4 will be incorporated into CAVA in CATIA V5.

This program is sold by TransCAT.

<http://www.transcat-plm.com/>

Regulations and internal agreements of the Volkswagen Group remain binding.

CAD-related applications - Creo Elements/Pro - applications

- Modpar
Database-supported editing and maintenance of CAD master data (Creo Elements/Pro standard parameters such as material, part number, associated standards). Furthermore, Modpar can be used to edit the revision record. In drawing mode, the revision record can be loaded from the model. A revision record included in the drawing can be imported into the model using Modpar.
- Pmap
Completion of Volkswagen standard parameters in Creo Elements/Pro
 - Amendment of pre-defined values, e.g., Creo Elements/Pro version
 - Transfer of old parameters, e.g., DESIGNATION to NAMING
 - Parameter designates for further database transactions
 - Automatic execution during model saving.
- ProRepair
Repair of faulty models (legacy data, data from external partners)
 - Repairs formal faults, such as missing layers, parameters, views
 - Automatic processing of recurrent repair work
 - Processing of whole assemblies
 - Processing of complete directories in batch mode (Windows)
- VW_ModelCHECK
This is used to initiate a check analogous to the check in KVS. After installation, the entry VW ModelCHECK (for KVS) can be found under menu item Analysis. For drawings, the query "ModelCHECK - Run on all drawing models? [N]" is answered automatically with "Y"; "All Levels" is always used for assemblies. The check is performed with regeneration. Subsequently, a window containing a summary of the test results of all sub-components is displayed. This report can be stored as a pdf file, printed, and used as transfer protocol.

The aforementioned Creo Elements/Pro extensions and others are included in the supplier package. This is provided free of charge on the B2B supplier platform, under Research and Development, 'Creo Elements/Pro', 'Download.'

Toolkit applications: Modpar, Pmap, ProRepair, and VW_ModelCHECK



Mr. Wolfgang Kunnert

Phone: +49 5361 896-3619

E-mail: wolfgang.kunnert@volkswagen.de

FuBIT

FuBIT is an application for the Creo Elements/Pro CAD/CAM system, which is used to generate and dimension holes and surfaces on complex geometries, such as powertrain parts, in a production-friendly manner. Dimensioning is carried out in tables, with the holes and surfaces sorted by machining sides and numbers. This means that these dimensions are not required on the drawing, which increases the clarity of the drawing, improves CAD model quality, and shortens turnaround times in the further course of the process chain, for example in NC programming. FuBIT allows the Creo Elements/Pro design engineer to access predefined "User Defined Features" (UDF). "User Defined Features" represent holes, which can be easily positioned on the 3-D model.

FuBIT can currently be used for Volkswagen Group orders and projects free of charge. FuBIT is included in the supplier package.

FuBIT



Mr. Thomas Reineke

Phone: +49 5361 896-3614

E-mail: Thomas.Reineke@volkswagen.de

- **Custom Animation (CA)**

The 'Custom Animation' application has been developed primarily for the transfer and visualization of movements that are not proprietary to Creo Elements/Pro. Proprietary to Creo Elements/Pro in this context refers to the kinematic tools MDX (Mechanism Design Extension) and Pro/MECHANICA Motion.

CA should be used for all externally determined movements, such as numerical simulations or measurement results from test stands.

Custom Animation is included in the supplier package.

CAD-related applications -DoLittle

DoLittle for CATIA and Creo Elements/Pro

DoLittle is an application integrated into the respective CAD environment that facilitates online creation of bilingual drawings by the EKDD/5 translation team.

Volkswagen achieves important benefits from use of DoLittle:

- **Immediate availability** of drawings that can be used internationally
- 90% reuse of standardized content with **high quality**
- **leading to lower translation/communication costs**
- International manufacturing also means language diversity. Volkswagen uses DoLittle to provide a standardized language throughout CAD projects.
- **Group standard VW 01058 is complied with, since it explicitly specifies use of DoLittle here.**

DoLittle is available for the applications CATIA V4, CATIA V5, and Creo Elements/Pro.

Further highlights:

- The central database currently holds approximately 85 000 standardized, bilingual drawing texts.
- Design engineers can search easily through the database and insert bilingual texts into the model.
- New texts may also be requested online.
- Requested texts are translated quickly by technical translators.
- The translated texts are stored in the database and available immediately.

Variable text parts (e.g., numerical data) increases reusability of the texts.

The current DoLittle version supports the language pairs
DE => EN and EN => DE

Important: DoLittle is not a machine translation tool!

This application can be obtained from T-Systems Enterprise Services GmbH ' (<https://servicenet.t-systems.de/>) subject to a fee.

DoLittle	
 (Translation team) Ms. Ulrike Kavemann	Phone: +49 5361 9-27460 E-mail: ulrike.kavemann@volkswagen.de
 (CATIA) Mr. Carsten Carulli	Phone: +49 5361 896-3617 E-mail: carsten.carulli@volkswagen.de
T-Systems International GmbH	Phone: +49 375 60619-901 E-mail: cax-service@t-systems.com
 (Creo Elements/Pro) Mr. Gerhard Kroschewski	Phone: +49 5361 896-3644 E-mail: gerhard.kroschewski@volkswagen.de

14. Engineering Portal

Process support for cooperative vehicle development

Modern vehicle development requires effective cooperation between specialists all over the world. The objective of an Engineering Portal is to provide a reliable platform for the organization of and collaboration between all engineering partners involved in the product emergence process. For this purpose, a common Engineering Portal has been set up for the Audi and Volkswagen Design Engineering divisions. Actual Audi and Volkswagen vehicle projects were the defining parameters in designing the portal. Modern portal technology provides online access to information and systems. All partners – both internal and external – access the current data around the clock. The Engineering Portal acts as a 'Home for Projects' and is intended as the binding access point to information and systems during the course of vehicle development.

Volkswagen Engineering Portal

The Engineering Portal (ePortal) allows users to upload and download current information and Office documents for project processing both from within the Group and externally, subject to an authorization concept. The central Engineering Portal ensures the cross-system availability of information, depending on the task (role) and project membership; it ensures transparent team work and prevents redundancies, thereby increasing quality and adherence to deadlines in project work.

Collaboration with partners

The ePortal provides a path to information on close collaboration at https://eportal.wob.vw.vwg/ict_ep/web/zusammenarbeit-mit-partnern. It serves the R&D departments and all R&D development partners as a central access point to all information required for successful collaboration.

Engineering Portal	
Operational Services GmbH & Co. KG Service Support Center (SSC)	Phone: 0800 5877 877 (Germany) Phone: +49 375 60619-904 (international) Fax: +49 1805 721433-444 E-mail: csn.service@o-s.de
 Hotline e2e hotline	Phone: +49 5361 9-86929 E-mail: e2e.hotline@volkswagen.de
 Collaboration with partners Anke Purschke	Phone: +49 5361 9-41821 E-mail: Anke.Purschke@volkswagen.de

The English translation is believed to be accurate. In case of discrepancies, the German version is alone authoritative and controlling. Numerical notation according to ISO/IEC Directives, Part 2.

15. Abbreviations

CAD/CAM	Computer-Aided Design/Computer-Aided Manufacturing
CA	Custom Animation
CATIA	One of the two key CAD/CAM systems used at VOLKSWAGEN AG
CEG	CATIA Einsatz Gruppe (Application Group)
ComSecure	Free solution for the encryption of data; available for various platforms
Creo/Elements Pro	Creo/Elements Pro (formerly ProENGINEER), one of the two key CAD/CAM systems used at VOLKSWAGEN AG
CSN	CAX Supplier Network
CUTNET	Application for the generation and representation of sections and grids in CATIA
DMU	Digital Mock-Up, virtual model
EDM	Engineering Data Manager
eM-Planner	Electronic Manufacturing Planner
ENDCHK	Checks CATIA models for compliance with VW 01059
ENGDAT	Engineering Data, CAD data exchange with at least 2 files: CAD file(s) and description file
ENGPART	Engineering Partner Data
ENX	European Network eXchange. A secure high performance network with a guaranteed and controlled service, to support the increasing demand for quicker development and cooperation between partners in the automotive industry beyond national borders.
FE	Research & Development
FTP	File Transfer Protocol, protocol used for transferring files
FuBit	Surface and hole information tables
GKV	Reach circles as per DIN ISO 3958; used for numerical simulations and representation of enveloping surfaces within hands' reach and their intersection curves
GURT	Belt anchoring areas
gzip	Compression software
HLS 4 ^{plus}	Web-based solution for company-wide management of documents, facilities, and inventories in factory planning
HyperKVS	Engineering data management system (Volkswagen system)
ICEM/SURF	Free form surface software
IGES	Initial Graphics Exchange Specification
INSI	Indirect view through a rear-view mirror; used to determine the minimum sizes for interior and exterior mirrors
IP address	Internet protocol address; unique address used to identify a computer within a network
ISDN	Integrated Services Digital Network
IUA	Provides CATIA users with functions to simplify tasks during development and design
ModelCHECK	Test tool used to check the quality of Creo Elements/Pro data
MODPAR	Database-supported completion of model parameters
NETLINE	Provides CATIA users with a function for creating grid lines in the Draw area or Space area

Abbreviations

ODETTE	Organization for Data Exchange by TeleTransmission in Europe; amalgamation of national, European associations from the automotive industry for the purpose of electronic data exchange
OFTP	Odette File Transfer Protocol
PDAs	Product data types in HyperKVS
PDM	Product Data Manager
PEP	Product Emergence Process
PMAP	Parameter Mapper
PUPPE	GII application for the generation of elements for ergonomic studies of an SAE dummy
RVS	Computer group system; communication monitor
SE	Simultaneous Engineering; used for quasi-parallelized engineering
SFV	Standard dedicated connection
STEP	Standard for the Exchange of Product Model Data
SWPII	GII application for the definition of welded joints in CATIA
TCP/IP	TCP/IP (Transmission control protocol/internet protocol) is a standard protocol, which is installed, for example, in local networks or on the internet
TDM	Team Data Manager
TIFF	Tagged Image File Format; industrial standard for raster data
TLS	Transport Layer Security
VALIDAT	Design data test program to determine the design-model data quality of elements and groups of elements as per defined test criteria
VDA	German Association of the Automotive Industry
VDAFS	VDA surface interface
VPM	Team Data Manager for the CATIA CAD system
WinZip	Windows compression program for compressing and decompressing data

16. Address list

Registration for connection set-up	
Operational Services GmbH & Co. KG Service Support Center (SSC)	Phone: 0800 5877 877 (Germany) Phone: +49 375 60619-904 (international) Fax: +49 1805 721433-444 E-mail: csn.service@o-s.de
 Mr. Christian Pomar (T-Systems)	Phone: +34 93 773-3151 E-mail: christian.pomar@t-systems.es
 T-Systems Czech Republic, a.s.	Phone: +420 326 71-1444 E-mail: KVS@t-systems.cz
 Mr. David Romero (T-Systems)	Phone: +34 93 773-1712 E-mail: extern.david.romero@seat.es
CAD/CAM methodology	
 Mr. Martin Ahlers	Phone: +49 5361 8-963612 E-mail: martin.ahlers@volkswagen.de
 Mr. Leandro Mouzo (T-Systems)	Phone: +34 93 774-4471 E-mail: extern.leandro.mouzo@seat.es
 Mr. Luboš Patka	Phone: +420 326 81-7316 E-mail: lubos.patka@skoda-auto.cz
 Research & Development Mr. Radek Ježek	Phone: +420 326 81-5328 E-mail: radek.jezek@skoda-auto.cz
CAD Interface Laboratory	
 CAD Interface Laboratory Mr. Steffen Hohmann	Phone: +49 5361 9-76589 E-mail: interfacelaboratory@volkswagen.de

CATIA V4/GII	
 T-Systems International GmbH	Phone: +49 375 60619-901 E-mail: cax-service@t-systems.com
 Mr. Miguel Delgado (T-Systems)	Phone: +34 93 773-3250 E-mail: extern.miguel.delgado2@seat.es

CATIA checking tools	
 Mr. Walter Pohlabein	Phone: +49 5361 896-3621 E-mail: walter.pohlabein@volkswagen.de
 Mr. Leandro Mouzo (T-Systems)	Phone: +34 93 774-4471 E-mail: extern.leandro.mouzo@seat.es

CATIA V4 standards	
 Mr. Michael Ditschun	Phone: +49 5361 896-3616 E-mail: michael.ditschun@volkswagen.de
 Mr. Leandro Mouzo (T-Systems)	Phone: +34 93 774-4471 E-mail: extern.leandro.mouzo@seat.es
 Mr. Stanislav Sedláček	Phone: +420 326 80-7005 E-mail: stanislav.sedlacek@skoda-auto.cz

CATIA V5	
 Mr. Michael Ditschun	Phone: +49 5361 896-3616 E-mail: michael.ditschun@volkswagen.de
 Mr. Martin Ahlers	Phone: +49 5361 896-3612 E-mail: martin.ahlers@volkswagen.de
 Mr. Leandro Mouzo (T-Systems)	Phone: +34 93 774-4471 E-mail: extern.leandro.mouzo@seat.es
 Mr. Stanislav Sedláček	Phone: +420 326 81-7005 E-mail: stanislav.sedlacek@skoda-auto.cz

CATIA V5/CAA	
 T-Systems International GmbH	Phone: +49 180 572 143 3510 E-mail: cax-service@t-systems.com
 Mr. Miguel Delgado (T-Systems)	Phone: +34 93 773-3250 E-mail: extern.miguel.delgado2@seat.es

Creo Elements/Pro checking tools (ModelCHECK)	
 Mr. Wolfgang Kunnert	Phone: +49 5361 896-3619 E-mail: wolfgang.kunnert@volkswagen.de
 Mr. Francesc Mestres (T-Systems)	Phone: +34 93 774-4762 E-mail: extern.francesc.mestres@seat.es
 T-Systems Czech Republic a.s. Mr. David Solín	Phone: +420 326 71-1410 E-mail: solin@t-systems.cz
 Mr. Jan Dvořák	Phone: +420 326 81-8923 E-mail: jan.dvorak@skoda-auto.cz

Digital factory	
 Mr. Frank Jelich	Phone: +49 5361 9-37862 E-mail: frank.jelich@volkswagen.de
 Mr. Uwe Koch	Phone: +49 5361 9-40644 E-mail: uwe.koch2@volkswagen.de
 Mr. Jürgen Kunhenn	Phone: +49 5361 9-39577 E-mail: juergen.kunhenn@volkswagen.de
 Mr. Jordi Zapatero Baro	Phone: +34 93 708-7176 E-mail: jordi.zapatero@seat.es
 Mr. Leoš Cervený	Phone: +420 326 81-9439 E-mail: leos.cervený@skoda-auto.cz

DMU	
 Mr. Christian Kutscha	Phone: +49 5361 9-38933 E-mail: christian.kutscha@volkswagen.de
 Mr. Blas Asensio Torres	Phone: +34 93 708-7983 E-mail: blas.asensio@seat.es
 Mr. Martin Kmínek	Phone: +420 326 81-5646 E-mail: martin.kminek@skoda-auto.cz

DoLittle	
 (Translation team) Ms. Ulrike Kavemann	Phone: +49 5361 9-27460 E-mail: ulrike.kavemann@volkswagen.de
 (CATIA) Mr. Carsten Carulli	Phone: +49 5361 896-3617 E-mail: carsten.carulli@volkswagen.de
T-Systems International GmbH	Phone: +49 375 60619-901 E-mail: cax-service@t-systems.com
 (Creo Elements/Pro) Mr. Gerhard Kroschewski	Phone: +49 5361 896-3644 E-mail: gerhard.kroschewski@volkswagen.de

Engineering Portal	
Operational Services GmbH & Co. KG Service Support Center (SSC)	Phone: 0800 5877 877 (Germany) Phone: +49 375 60619-904 (international) Fax: +49 1805 721433-444 E-mail: csn.service@o-s.de
 Hotline e2e hotline	Phone: +49 5361 9-86929 E-mail: e2e.hotline@volkswagen.de
 Collaboration with partners Anke Purschke	Phone: +49 5361 9-41821 E-mail: Anke.Purschke@volkswagen.de

ENGPART import data

KVS service team

Phone from Germany: 0800 5 877 877

Phone from outside Germany: +49 375 354 28978

E-mail: csn.service@o-s.de**FuBIT**

Mr. Thomas Reineke

Phone: +49 5361 896-3614

E-mail: Thomas.Reineke@volkswagen.de**HLS**

User support

HLS hotline

Phone: +49 5361 9-46192



Design Engineering

Mr. Horst Röhl

Phone: +49 5361 896-3930

E-mail: horst.roehl@volkswagen.de

Volkswagen support

Mr. Rainer Koslik

Phone: +49 5361 9-20977

E-mail: rainer.koslik@volkswagen.de

Skoda support

Mr. Leoš Červený

Phone: +420 326 81-9439

E-mail: leos.cerveny@skoda-auto.cz

SEAT support

Mr. Jordi Zapatero Baro

Phone: +34 93 708-7176

E-mail: jordi.zapatero@seat.es

ASIC support

Dr. Ommo Ommen

Phone: +49 5361 9-25082

E-mail: extern.ommo.ommen@volkswagen.de

HyperKVS: Administrative queries, contract processing, technical connection

Operational Services GmbH & Co. KG Service Support Center (SSC)	Phone: 0800 5877 877 (Germany) Phone: +49 375 60619-904 (international) Fax: +49 1805 721433-444 E-mail: csn.service@o-s.de
 KVS-SEAT hotline	Phone: +34 93 773-1091 (application) Phone: +34 93 773-3298 (network) E-mail: kvs@seat.es
T-Systems Czech Republic a.s. Hotline	Phone: +420 326 71-1445 Fax: +420 326 71-1420 E-mail: kvs@t-systems.cz

ENDCHK, VALIDAT installation support (subject to a fee)

T-Systems International GmbH	Phone: +49 375 60619-901 E-mail: cax-service@t-systems.com
 T-Systems Czech Republic a.s. Mr. David Solín	Phone: +420 326 71-1410 E-mail: solin@t-systems.cz
 Mr. Leandro Mouzo (T-Systems)	Phone: +34 93 774-4471 E-mail: extern.leandro.mouzo@seat.es

Kate

 Mr. Heimo Lütge	Phone: +49 5361 611-39622 E-mail: heimo.luetge@volkswagen.de
 Mr. Jordi Zapatero Baro	Phone: +34 93 708-7176 E-mail: jordi.zapatero@seat.es
 Mr. Leos Červený	Phone: +420 326 81-9439 E-mail: leos.cervený@skoda-auto.cz

E-mail encryption

Operational Services GmbH & Co. KG Automotive Desk Rudolf-Ehrlich-Str. 7 D-08058 Zwickau	Phone: +49 375 60619-903 (Germany) Phone: +49 375 60619-904 (international) Fax: +49 1805 231255 E-mail: servicesupport.center@o-s.de
---	--

Standards	
 Mr. Norbert Wisla	Phone: +49 5361 9-48869 E-mail: norbert.wisla@volkswagen.de

NVS (standard parts accessible to partner companies)	
Operational Services GmbH & Co.KG Service Support Center (SSC)	Phone 0800 5 877 877 (Germany) Phone: +49 375 60619-904 (international) Fax: +49 1805 721433-444 E-mail: csn.service@o-s.de

OFTP	
 Mr. Bodo Meier	Phone +49 5361 9-76818 E-mail: bodo.meier@volkswagen.de
 Mr. Daniel Gregor	Phone: +420 326 81-7463 E-mail: daniel.gregor@skoda-auto.cz

Plant Simulation	
 Mr. Matthias Schwarz	Phone: +49 5361 9-72520 E-mail: matthias.schwarz@volkswagen.de
 Mr. Carsten Pöge	Phone: +49 5361 9-39327 E-mail: carsten.poege@volkswagen.de
 Mr. Leoš Červený	Phone: +420 326 81-9439 E-mail: leos.cerveny@skoda-auto.cz
 Mr. Carlos Banus Giro	Phone: +34 93 708-5058 E-mail: carlos.banus@seat.es

Plot data	
 CAD Interface Laboratory Mr. Steffen Hohmann	Phone: +49 5361 9-76589 E-mail: interfacelaboratory@volkswagen.de
 Mr. Rainer Klinger	Phone: +49 5361 9-76567 E-mail: rainer.klinger@volkswagen.de
 Mr. Xavier Calafí (T-Systems)	Phone: +34 93 773-3297 E-mail: francesc.calafi@seat.es
 Mr. Marek Dvořák	Phone: +420 326 81-8792 E-mail: marek.dvorak@skoda-auto.cz

Process Designer	
 Mr. Uwe Zaretzke	Phone: +49 5361 9-196584 E-mail: uwe.zaretzke@volkswagen.de
 Mr. Gunner Jensen	Phone: +49 5361 9 40768 E-mail: Gunner.Jensen@volkswagen.de
 Mr. Leoš Červený	Phone: +420 326 81-9439 E-mail: leos.cerveny@skoda-auto.cz
 Mr. Jordi Zapatero Baro	Phone: +34 93 708-7176 E-mail: jordi.zapaterp@seat.es

Robcad robot simulation and offline programming	
 Mr. Manfred Labersweiler	Phone: +49 5361 9-39481 E-mail: manfred.labersweiler@volkswagen.de
 Mr. Jordi Zapatero Baro	Phone: +34 93 708-7176 E-mail: jordi.zapatero@seat.es
 Mr. Leoš Červený	Phone: +420 326 81-9439 E-mail: leos.cerveny@skoda-auto.cz

'Provision of CAD/CAM data' service of the Volkswagen Group Research & Development Department

 R&D Data Exchange	Phone: +49 5361 9-78951 E-mail: FE-datenaustausch@volkswagen.de
 R&D Data Provision	Phone: +49 5361 9-29588 E-mail: FE-Informationsservice@volkswagen.de

Technical pilot data exchange

 CAD Interface Laboratory Mr. Steffen Hohmann	Phone: +49 5361 9-76589 E-mail: interfacelaboratory@volkswagen.de
 Mr. Leandro Mouzo (T-Systems)	Phone: +34 93 774-4471 E-mail: extern.leandro.mouzo@seat.es

Toolkit applications: Modpar, Pmap, ProRepair, and VW_ModelCHECK

 Mr. Wolfgang Kunnert	Phone: +49 5361 896-3619 E-mail: wolfgang.kunnert@volkswagen.de
--	--

Connection variants and bandwidth

Operational Services GmbH & Co. KG Service Support Center (SSC)	Phone: 0800 5877 877 (Germany) Phone: +49 375 60619-904 (international) Fax: +49 1805 721433-444 E-mail: csn.service@o-s.de
 Mr. Christian Pomar (T-Systems)	Phone: +34 93 773-3151 E-mail: christian.pomar@t-systems.es
 Mr. David Romero (T-Systems)	Phone: +34 93 773-1712 E-mail: david.romero@seat.es

Procedural instructions

 Mr. Stefan Biernoth	Phone: +49 5361 9-46547 E-mail: stefan.biernoth@volkswagen.de
 Mr. Miguel Montano Ibañez	Phone: +34 93 708-7057 E-mail: miguel.montano@seat.es

VPM	
 Mr. Torsten Pürschel	Phone: +49 5361 896-3839 E-mail: torsten.puerschel@volkswagen.de
 Mr. Guido Fischer	Phone: +49 5361 9-35622 E-mail: guido.fischer@volkswagen.de
 Mr. Raul Gabas Koeppel (T-Systems)	Phone: +34 93 708-7041 E-mail: raul.gabas@seat.es

Other applications in the partner company network	
 Mr. Heiko Rolof	Phone: +49 5361 9-49737 E-mail: heiko.rolof@volkswagen.de
Operational Services GmbH & Co. KG Service Support Center (SSC)	Phone: 0800 5877 877 (Germany) Phone: +49 375 60619-904 (international) Fax: +49 1805 721433-444 E-mail: csn.service@o-s.de
 Mr. Christian Pomar (T-Systems)	Phone: +34 93 773-3151 Fax: +34 93 773-3252 E-mail: christian.pomar@seat.es
 Mr. David Romero (T-Systems)	Phone: +34 93 773-1712 E-mail: david.romero@seat.es